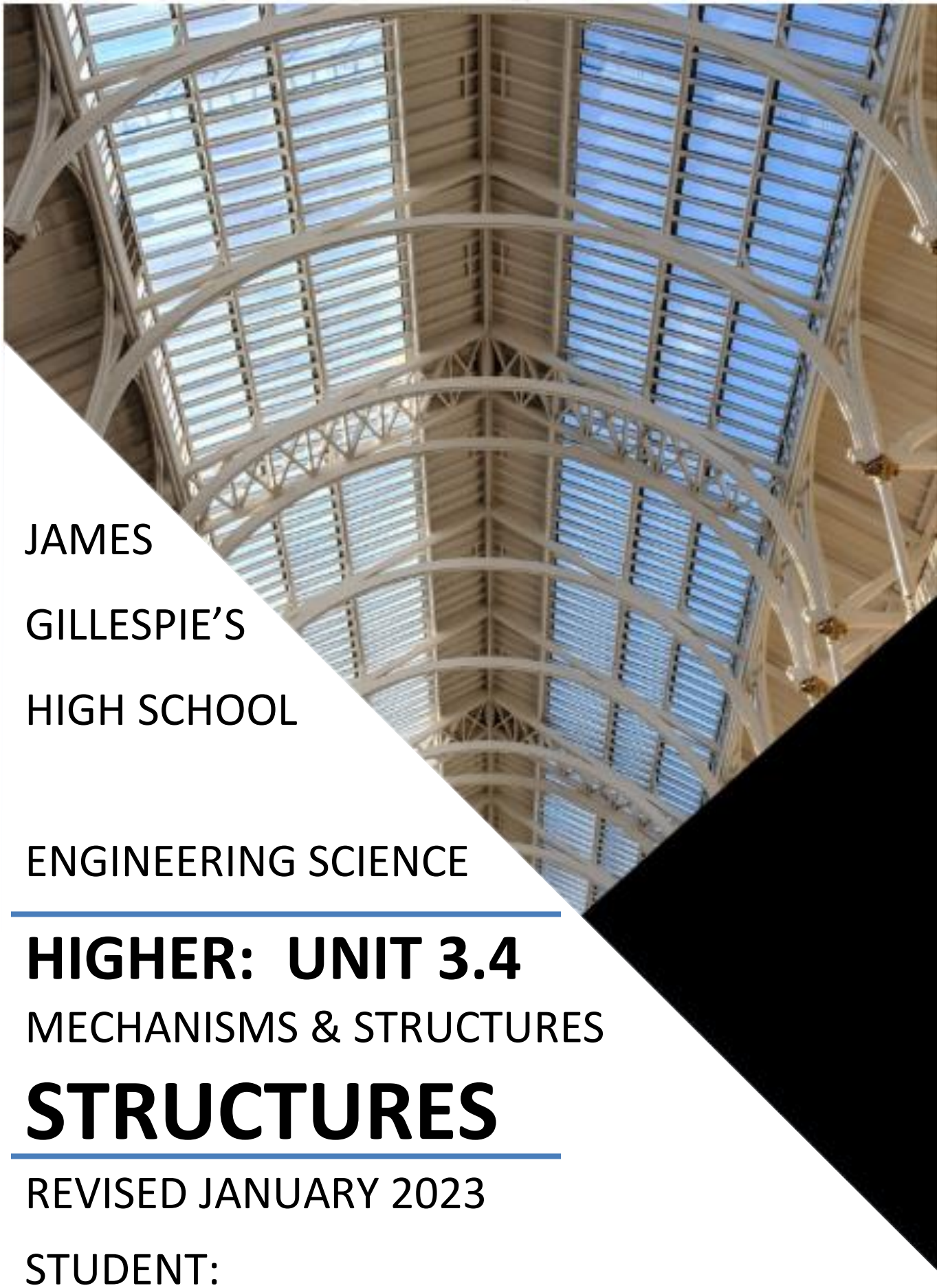


JAMES

GILLESPIE'S

HIGH SCHOOL

ENGINEERING SCIENCE

HIGHER: UNIT 3.4

MECHANISMS & STRUCTURES

STRUCTURES

REVISED JANUARY 2023

STUDENT:

YEAR: TEACHER:

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Outcomes and Assessment Standards

This unit will build on skills learned in National 5 Engineering Science in order to further deepen your understanding of Mechanisms and Structures to enable you to design your own working systems.

To make sure you have all the understanding needed to succeed at Higher Engineering Science, record your progress using the following colour codes:

Red: I need to work on this topic further to understand it.

Amber: I need a bit of help to fully understand this topic.

Green: I am confident in my understanding of this topic.

I can research and investigate a range of complex mechanisms and structures by:

- Using the systems approach to analyse mechanisms and structures.
- Describing or producing diagrams of a range of complex structures.
- Describing or producing diagrams of a range of complex mechanisms.
- Investigating the properties of a range of materials used in mechanisms.

Learning intentions

- Draw a free body diagrams for force systems.
- Recognize how ties and struts behave in a frame structure.
- Understand that a node in a frame structure is in equilibrium.
- Apply the conditions of static equilibrium at a frame structures node.
- Use the nodal analysis method to determine the loads in members.
- Be able to determine the support reactions for a frame structure.
- Apply the conditions of static equilibrium in solving problems on concurrent force and non-concurrent force systems.
- Resolve a force into its horizontal and vertical components.
- Find the resultant force in concurrent force system.
- Apply the principle of moments to a simple lever system.
- Apply the conditions of static equilibrium to solve non-concurrent force systems.

I can develop mechanical or structural solutions to solve complex problems by:

- ☐ Identifying key aspects of the problem.
- ☐ Applying knowledge and understanding of structures, materials and/ or mechanisms.
- ☐ Carrying out calculations to assist the selection of materials or component sizes.
- ☐ Designing structures and / or mechanisms.
- ☐ Simulating or building mechanisms and/ or structures.
- ☐ Testing and evaluating solutions.
- ☐ Testing and evaluating programs against a specification.

Structures

It is very important with any structure that we can calculate the forces acting within it so that a safe structure can be designed. Early structures were found to be successful due to the fact that they stayed up and many early structures are still with us, but many are not. The science of structures has been progressively improving over the centuries and it is now possible to predict structures behaviour by analysis and calculation. Errors can still be made, sometimes with catastrophic results.

In this course we are generally going to consider frame structures. We shall learn to recognise the effect a load will have on a body and what happens if several forces are acting on a structure. When a force acts on a body we have to be able to calculate the effect on the different members. We shall by the end of this unit be able to calculate the force acting in any member in a simple structure and, from this determine, a suitable material and cross section needed to carry this load.

What is a structure?

A structure is an arrangement of materials or a combination of material which is designed to support a load in some way. Structures are all around us and can either be man-made or natural. Trees and bee hive honeycombs are great examples of natural structures while construction cranes and bridges are man-made. A successful structure is one which has sufficient strength to carry out its task without being too heavy or use to much material. Good structures depend upon:

1. The strength of the material used;
2. The form and arrangement of all the different parts of the structure;
3. The design and quality of the joining methods.

Mass

Mass is the amount of matter in a body and ais a measure in kilograms (KG). The mass of a body does not change, no matter where you take it. If you take 1kg of sugar from Earth to the moon, you would still get the same amount of tea spoons of sugar for your tea.

Weight

The weight of an object however does change. Weight is the gravitational pull on a body. Weight is measured in Newtons (N). On Earth, the gravitational pull on a mass of 1kg is 9.81N. Therefore if you want to find the weight of an abject on Earth, you multiply its mass by 9.81. We only use 9.8N as stated in the Engineering Science data booklet.

Force

Force is a vector quantity, and as such must be represented in Magnitude and Direction. Forces tens to cause motion in the direction that they are acting. Force is measured in Newtons (N). Forces can be represented by straight lines with an arrow head to show the direction they are acting. In the example below, the total of the two forces acting towards the block is 300N but there is an opposing force of 50N on the other side, possible due to friction which must be overcome. The ‘Resultant force’ is 250N acting on the block and therefore it will move from left to right.

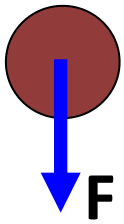


Vectors

All structures are loaded or exerted upon by external forces. The many different physical quantities you will encounter in the field of engineering and science may be split into two different groups. The first group contains those quantities that have no direction; e.g. mass and volume are examples of scalar quantities.

The second group of physical quantity posses a direction as well as a magnitude and are known, as a vector quantity examples of vector quantities are displacement, velocity and force.

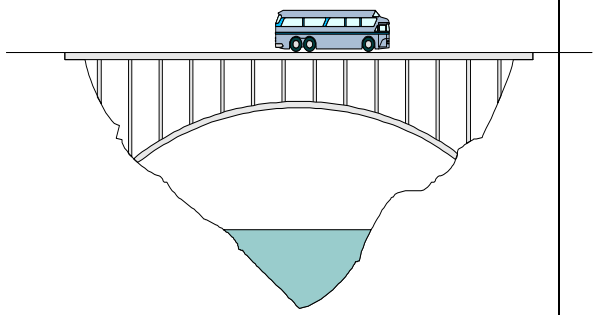
A vector quantity can be represented by a straight line drawn in the direction its acting. The length of the line is proportional to the magnitude. The diagram below shows the vector for the force produced by gravity acting on a 10 kg force.



F= Mass (m) x Gravitational field strength (g) therefore F = mg
F = 10 x 9.8
F= 98N

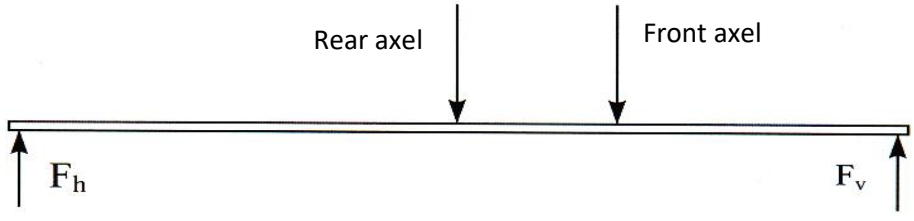
Free-body diagrams

It is important to isolate different parts of a structure or body from its adjacent surroundings. In a line diagram this can be done by drawing a free-body diagram, which is a diagrammatic representation of all or part of the structure showing the forces affecting it.



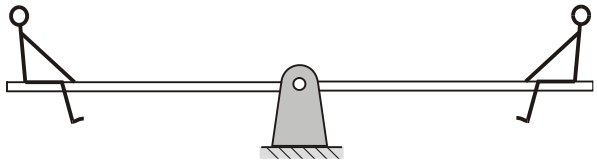
If all the visual components acting on the structure or body were removed and replaced with their force value, a simplified diagram would allow a better understanding of how the forces are affecting the structure.

The diagram below is a simplified free-body diagram of the bridge loaded by the bus. The forces representing the bus and the weight of the bridge act downwards and are taken through the centre of the bus and the middle of the bridge. Because of the point of contact of the bus, the arrow is drawn through its centre. The forces F_h and F_v represent the forces that the supports have on the structure, therefore they also have to be shown. We could be more detailed and draw the angled support for the bridge in the rock face.

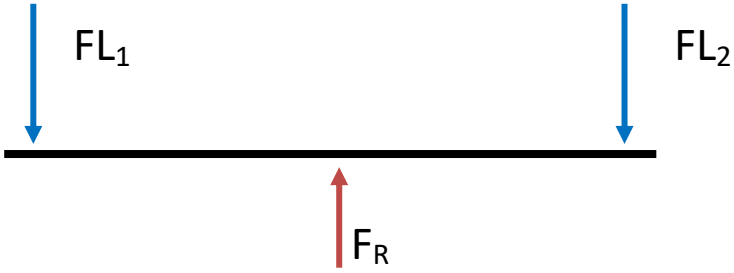


Adding vectors

A playground see-saw structure has three forces acting on it. F1 and F2 are forces acting due to the mass of the people and F3 is the reaction force to F1 and F2. The reaction force is calculated by adding together the downward forces. In any force system, the sum of the downward vertical forces must equal the sum of the upward vertical forces to create **eqilibrium**. $\Sigma Fv = 0$



The free body diagram shown below can represent the forces in the above diagram.



FL_1 and **FL_2** represent the force exerted due to the mass (Load) of the people. **F_R** is the reaction force. The reaction load **F_R** is found by adding the two downward forces together.

In any force system the sum of the vertical forces must be equal to zero.

$$\Sigma F_v = 0$$

$$F_R = FL_1 + FL_2$$

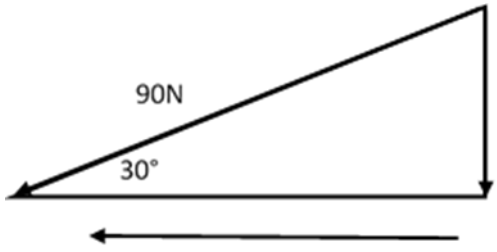
$$F_R = 810N + 740N$$

$$F_R = 1550N$$

$\Sigma F_R = 0$ - This is one of the conditions of static equilibrium.

There are two other conditions of static equilibrium:

1. All forces acting horizontally must be = to 0
2. ΣCWM = ΣACWM states that clockwise forces = anticlockwise forces.



Types of structure: mass, framed and shells.

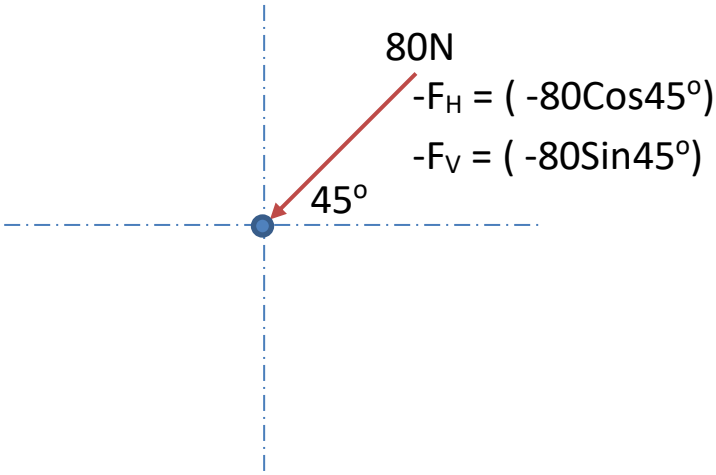
1. Mass structures perform due to their own weight. An example would be a dam.

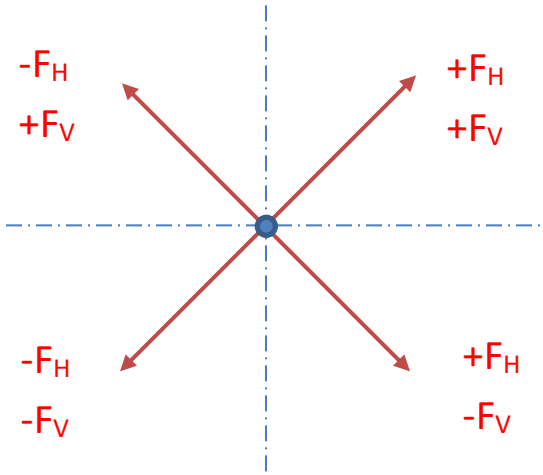
2. Frame structures resist loads due to the arrangement of its members. A house roof truss can support a load many times its own weight. A two dimensional frame such as this is known as a plane frame. The Eiffel Tower is an example of a three dimensional frame structure, known as a space frame. Electricity pylons are good examples of frame structures.

3. Shells are structures where its strength comes from the formation of sheets to give strength. A car body is an example of a shell structure.

The Resultant

The resultant is the vector sum of two or more vectors (start point to end point). It is basically a single force which demonstrates the sum (total) of all the forces acting on an object. To calculate the resultant when various angles are acting on an object, you need to simplify those acting forces to just horizontal and vertical forces. If the direction of the force is pointing down and to the left, this is a double negative number (-Fv and $-F_H$). Think of creating a graph as shown below of x and y axes.

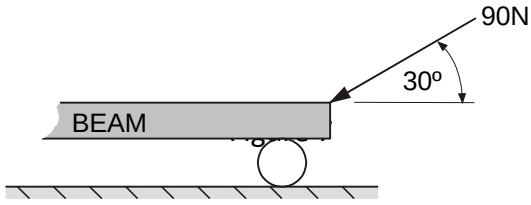




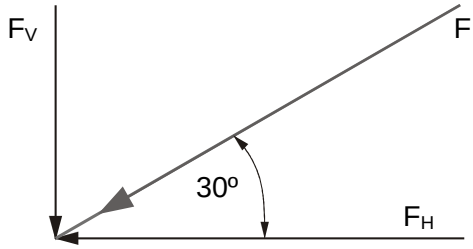
Delivered via Powerpoint notes.

Structures and Forces:
Resolution of forces (forces at angles)

In some cases, the loads may not be acting in the same direction and cannot therefore be added together directly. In the situation shown below the force is acting down at an angle.



This force can be split into two separate components. A vertical component **F_V** and a horizontal component **F_H**.

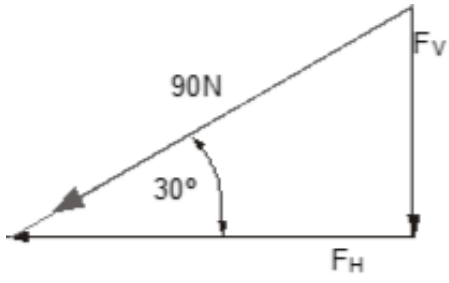


To resolve a force into its components, you will have to know two things; its **magnitude** and **direction**. Trigonometry is used to resolve forces. For this topic it is assumed that students have no previous experience with force systems. It would, however, be useful if the students had a working knowledge of basic trigonometry and can find an unknown dimension or angle in a right angled triangle.

So/h Ca/h To/a **'SohCahToa'** **Sin a = opp / hyp** **Cos a = adj / hyp**

Where:
Hypotenuse = Force (F)
Opposite = Vertical component (F_V)
Adjacent = Horizontal component (F_H)

The diagram above can be redrawn as opposite:



To find the Horizontal force (F_H)

$$\cos 30° = \frac{F_H}{90}$$

$$F_H = 90 \times \cos 30°$$

$$= 77.94N$$

RULE:
Σ F_H = Force multiplied by Cos Angle°

To find the Vertical force (F_V)

$$\sin 30° = \frac{F_V}{90}$$

$$F_V = 90 \times \sin 30°$$

$$= 45N$$

RULE:
Σ F_V = Force multiplied by Sin Angle°

1. Resolve the following forces into **only** their horizontal and vertical components.

a) 6KN, 60°

b) 10N, 30°

c) 70°, 25KN

d) 200N, 40°

Combining forces

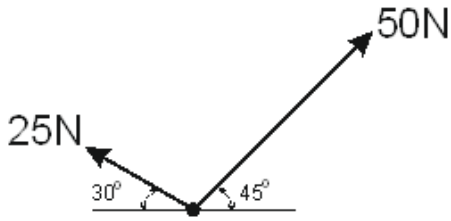
An object or structure will often have a number of different forces acting on it. By resolving the forces (adding them together) we can work out their combined effect. This can be done by force diagrams (shown below), as was covered at National 5 level, or by doing it mathematically.

What are Concurrent forces?

There can be many forces acting within a structure. Sometimes it is required to calculate the concurrent force to ensure static equilibrium is happening. It opposes all other forces by calculating the **magnitude** and **angle** of the force R required to **maintain equilibrium**.

Worked example of concurrent forces

A force diagram is quite a simple method but requires very accurate measurement to be effective. It becomes increasingly difficult as the number of forces increases.



Step 1: If the individual forces are resolved into vertical and horizontal components, they can be added together mathematically to get a more accurate result.

$\Sigma F_V = +25\text{Sin}30^\circ + 50\text{Sin}45^\circ =$ **47.8N** (Force Vertical)

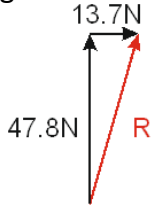
$\Sigma F_H = -25\text{Cos}30^\circ + 50\text{Cos}45^\circ =$ **13.7N** (Force Horizontal)

Step 2: Using Pythagoras' theorem, the magnitude of the **Resultant force** can be calculated.

$R = \sqrt{(A^2 + B^2)}$

$R = \sqrt{(F_V^2 + F_H^2)}$

$\mathbf{R} = \sqrt{(47.8^2 + 13.7^2)} =$ **49.7N**

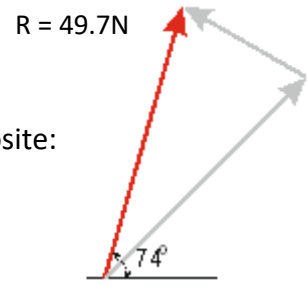


Step 3: The **direction** of the force can be calculated with trigonometry.

$\text{Angle} = \text{Tan}^{-1}(\Sigma F_V / \Sigma F_H)$

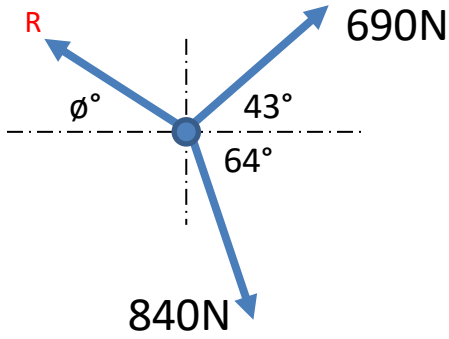
$\text{Angle} = \text{Tan}^{-1}(47.8 \div 13.7) =$ **74°**

Step 4: The results can be demonstrated in the diagram opposite:



Concurrent forces worked example 1 (pupil booklet)

A concurrent force system is shown.



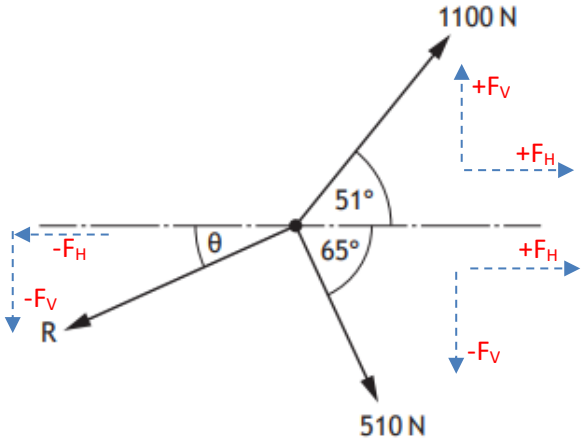
Calculate the magnitude and angle of the force R required to maintain equilibrium.

Answer on Powerpoint

Past paper SQA 2017 Q2

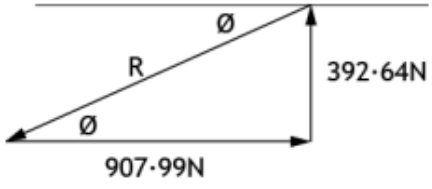
A concurrent force system is shown below.

Calculate the magnitude of the force R, and the angle θ, to maintain equilibrium.



It is useful to identify the Vertical (F_V) and Horizontal (F_H) forces at each pulling direction force. Is the concurrent force acting in a positive or negative direction at the point of Equilibrium?

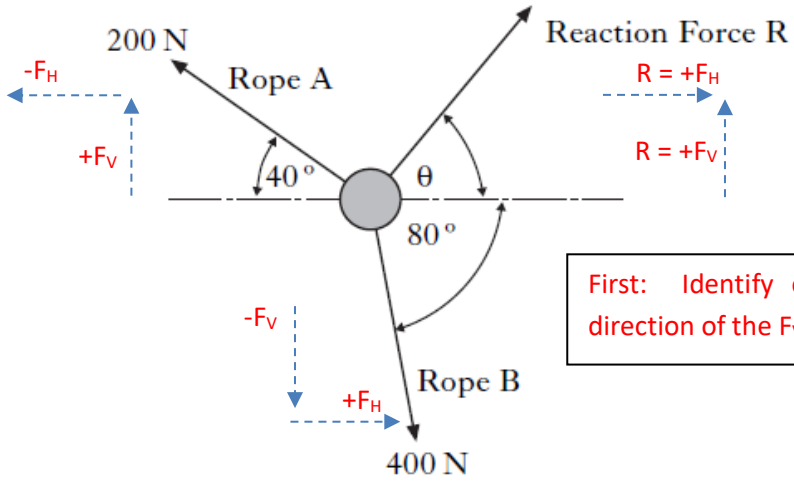
Answer

Expected answer(s)	Max mark	Additional guidance
$\Sigma F_V = 0$ $R_V = 1100 \sin 51 - 510 \sin 65 = 392.64$ N	4	1 mark for magnitude
$\Sigma F_H = 0$ $R_H = 1100 \cos 51 + 510 \cos 65 = 907.79$ N		1 mark for magnitude
		
$R = \sqrt{(392.64^2 + 907.99^2)} = 989$ N		1 mark for value with unit
Tan Θ = 392.64/907.99 => Θ = 23.4°		1 mark for value with unit

Concurrent forces worked example 2

Two boats are tied to a mooring post by polypropylene ropes.

The forces acting concurrently on the mooring post are shown in Figure Q3.



First: Identify each Force and the direction of the F_V and F_H.

Figure Q3

(*a*) Calculate the magnitude of the reaction force R, and the angle θ.

Each polypropylene rope has a diameter of 20 mm and a length of 20 m when unloaded.

(*b*) Calculate the change in length of rope B when loaded as shown in Figure Q3.

a)

$\Sigma F_V = 0$

$+RFv - 400NSin80° + 200NSin40° = 0$

$+RFv - 393.9231 + 128.5575 = 0$

$RFv = +\underline{265.366N}$

$\Sigma F_H = 0$

$+RF_H \ (+400NCos80° + (- 200NCos40°) = 0$

$+RF_H + 69.4593 - 153.2089 = 0$

$RF_H = \underline{+83.7496N}$

Calculation of magnitude: $R = \sqrt{ (F_V^2 + F_H^2)}$

$R = \sqrt{(265.366^2 + 83.7496^2)}$

$R = \sqrt{77428.3}$

$R = 278.26N \quad = \underline{280}$ (2s.f)

Angle $= Tan^{-1} (\Sigma F_V \div \Sigma F_H)$

Angle $= Tan^{-1} (265.366 \div 83.7496) =$ **72.5°** **(<u>73°</u>)**

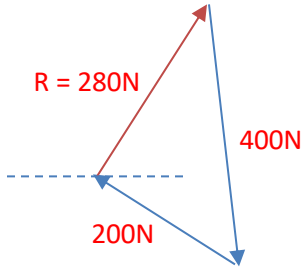
b)

$A = 314 \text{ mm}^2$

$\sigma = F/A$
$= 400/314$
$= 1.27 \text{ N/mm}^2$
$E = 0.9 \times 10^3 \text{ N/mm}^2$

$\varepsilon = \sigma/E$
$\varepsilon = 1.27/0.9 \times 10^3$
$= 0.00141$

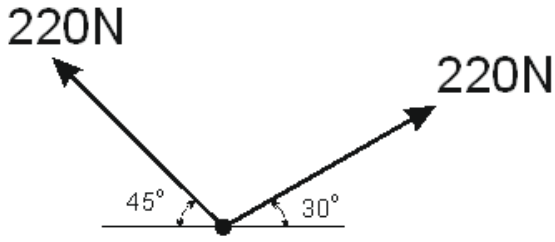
$\Delta l = \varepsilon \times 1$
$\Delta l = 0.00141 \times 20 \times 10^3$
$\Delta l = 28.2\text{mm}$



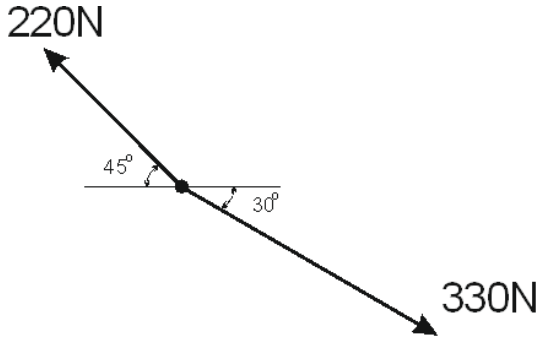
2.

Calculate the **magnitude** and **direction** of the resultant force in the following examples.

a)

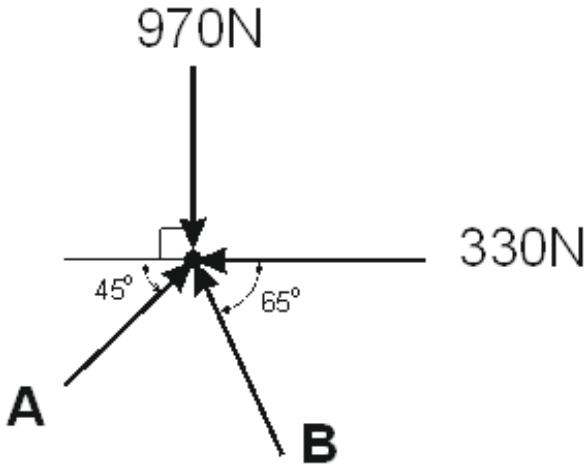


b)



c) A bike frame is being tested to ensure it can withstand the loads produced when involved in a downhill race. Using the information in the free-body diagram, calculate the forces inside members A and B.

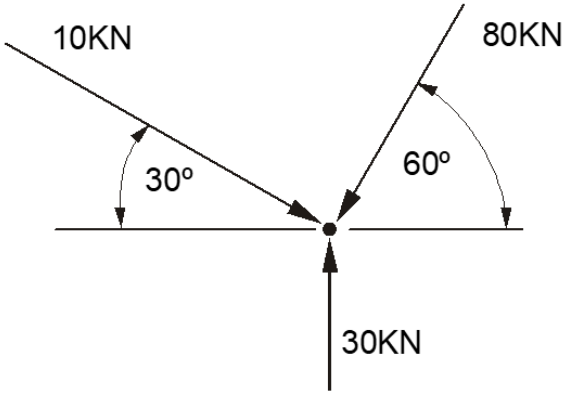




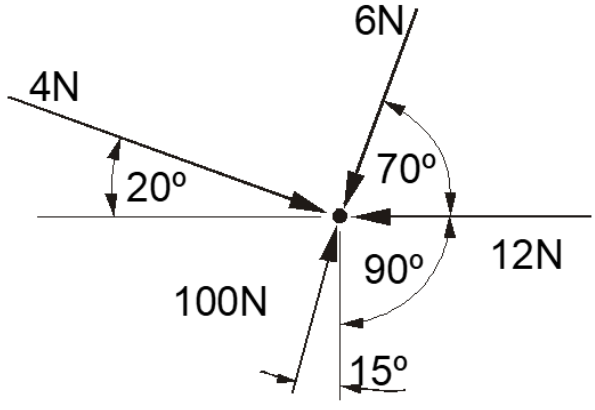
3.

Calculate the resultant for these force systems. Find the horizontal and vertical components of each - add them up to find the overall component forces and then find the resultant.

a)

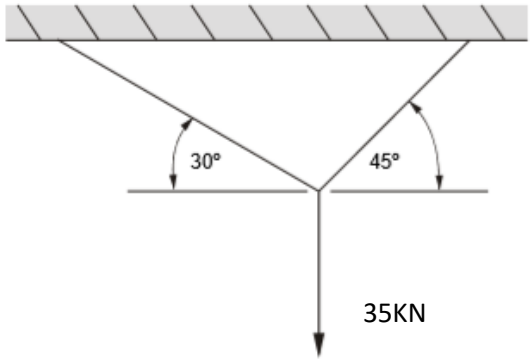


b)



Complex resolution of forces (Simultaneous equations)

Sometimes, when calculating the different parts if a structure, the components are known. Instead, it is the force that has to be calculated. This makes it far more difficult and may necessitate the use of simultaneous equations to work it out.



Sometimes the components are known and it is the force that is to be calculated. Determine the tension in the two cables. Note - the horizontal component of the two cables will be the same. (A simultaneous equation will be required)

$\Sigma F_H = 0$

$\Sigma \leftarrow = \Sigma \rightarrow$

$F_1 \times \text{Cos } 30^o = F_2 \times \text{Cos } 45^o$

$F_1 \times 0.866 = F_2 \times 0.707$

$0.866\ F_1 = 0.707\ F_2$ (Eqn 1)

$\Sigma F_V = 0$

$\Sigma \uparrow = \Sigma \downarrow$

$(F_1 \times \text{Sin } 30^o) + (F_2 \times \text{Sin} 45^o) = 35KN$

$0.5F_1 + 0.707F_2 = 35KN$ (Eqn 2)

From these two statements, simultaneous equations can be used. To resolve the missing force, transfer any relevant data from one equation into the other. It was first calculated that $0.866F_1 = 0.707F_2$. Therefore, this can be transferred into the second equation.

$0.5F_1 + 0.707F_2 = 35KN \rightarrow 0.5F_1 + 0.866F_1 = 35KN$

$1.37F_1 = 35KN$

$F_1 = 35KN \div 1.37 = \underline{25.5KN}$

Substitute value of F_1 back into Eqn 1:

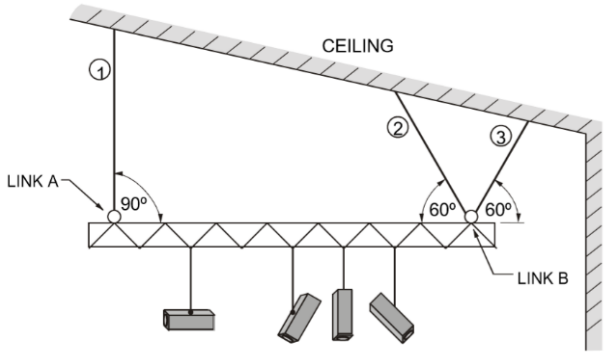
$0.866\ F_1 = 0.707\ F_2$

$0.866 \times 25.5KN = 0.707\ F_2$

$22.083 = 0.707F_2$

$F_2 = 22.083 \div 0.707 = \underline{31.23KN}$

4. A lighting gantry in a small theatre is fixed to a sloping ceiling by three independent wire ropes as shown below. When the lamps are set in position, the vertical loading on link A and link B are 2.5KN and 5KN respectively.



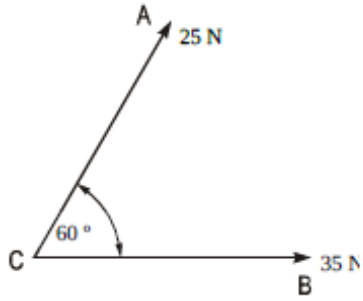
A) Calculate the magnitude of the force in each of the three ropes.

B) Recalculate the force in the ropes if the angle between rope 3 and the gantry changed to 45˚ instead of 60˚

Polygon of forces

A **vector diagram** is used to find the resultant of two forces that are inclined at an angle to each other.

Example 1



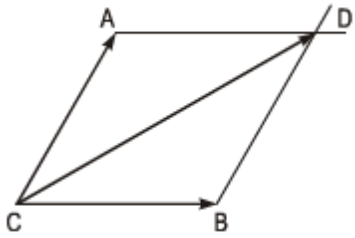
In the example above the **resultant** (the force which could replace the two given forces) of the two forces can be found by drawing two vectors. First choose a suitable scale and redraw the two vectors CA and CB at the scale of: 1 mm represents 1N.

25 mm = 25 N = CA

35 mm = 35 N = CB

The bigger the scale, the more accurate the vectors.

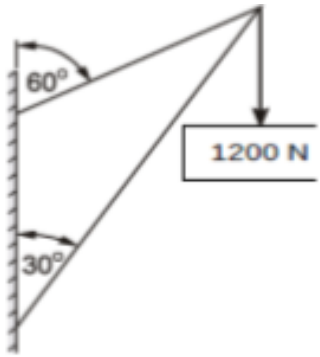
From the end point of A, draw a line which is parallel to line CB. From the end point of B draw, draw a line which is parallel to line CA. Identify where the two lines intersect as point D. Now draw a line from C to D. A line drawn from C to D is the resultant of the two forces CA and CB and can now be measured against the scale.



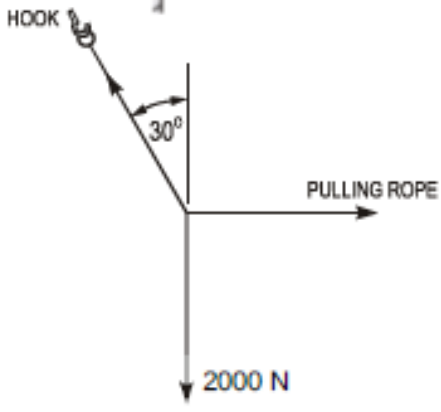
The **resultant** has a magnitude (size) of 46 N by scale measurement.

5.

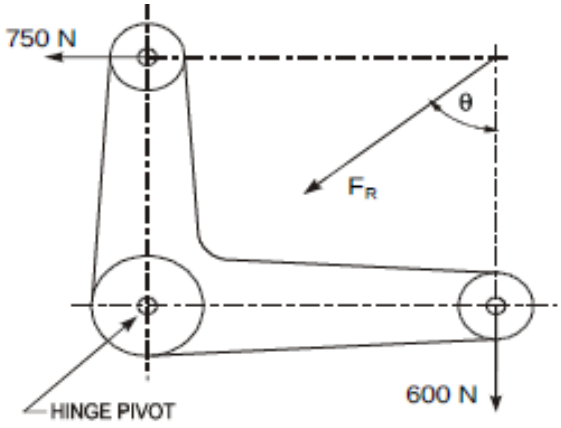
A) A small crane is used on a fishing trawler to lift cases of fish to the dock. The weight of the lift is 1200 N. Determine the size and direction of the forces in each of the crane members. (Use a scale where 10 mm represents 200 N.)



B) A weight of 2000 N is suspended by a rope attached to a hook firmly fixed to a roof joist. A second rope is attached to the vertical rope and pulled horizontally until the rope makes an angle of 30° to the vertical as shown. Determine the horizontal pull on the rope and the force on the hook

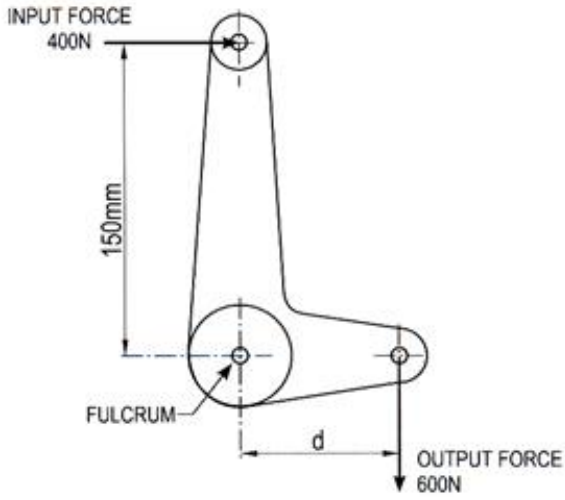


C) The figure below shows a cranked lever that is part of a gear-change mechanism. Find the resultant force FR acting on the hinge pivot and the angle θ.



Teacher worked examples:

Use the principles of moments to determine the length of the output side of the bell crank lever. Calculate the force multiplier ratio of the lever.



The principle of moments

The Greek letter ∑ (sigma) stands for 'the sum of' and can be used as a shorthand way of writing the principle of moments. It states that the 'sum of the Moments' (M) must equal zero (**ΣM = 0**) or the 'sum of the clockwise moments' must equal the 'sum of the anticlockwise moments' for a system to be in 'Equilibrium' (balanced and stationary).

1: Sum of ClockWise Moments = Sum of Anti ClockWise Moments

(∑CWM = ∑ACWM)

2: ∑Acting upwards Moments = ∑Acting downwards Moments

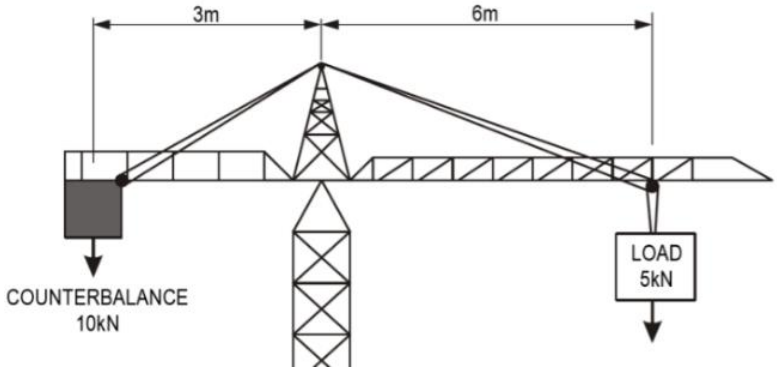
3: ∑Acting left Moments = ∑Acting right Moments

Moment = force (f) x distance (d)

$F_1 \times d_1 = F_2 \times d_2$

The 'turning effect' that can occur on a structure to due force is called a **moment**. A moment is measured in Newton metres (Nm). The moment is calculated by the force multiplied by the distance turning the lever clockwise is equal to the force multiplied by the distance turning the lever anticlockwise.

Large and complex mechanical systems, such as cranes, use moments to balance either side of the arm, otherwise the crane would topple over. When an item is perfectly balanced, it is called equilibrium. It is very important in the design of some structures to be in equilibrium or it will fail. The crane is in equilibrium as shown below:

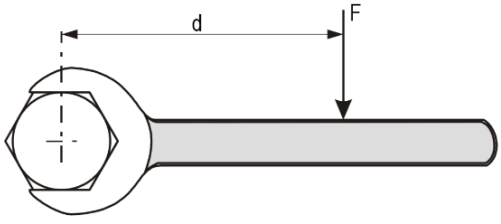


∑Anti Clockwise Moment = ∑Clockwise (load) Moment

10KN × 3m = 5KN x 6m

30 KNm = 30 KNm

Using a lever, or the rule of moments, you can apply a small force with a short distance from the pivot to produce a turning force. This effect is effective when used to tighten or loosen a bolt with a spanner.



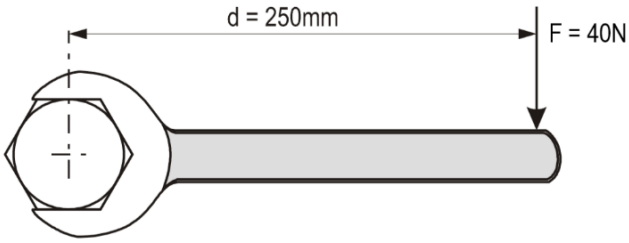
The effect of producing a large output force from a small input force is called a moment.

A larger moment is produced by moving the force further away from the pivot.

The moment of a force is dependent on the size of the force and the distance at which it is applied from the pivot point.

Worked example:

The spanner shown in the diagram is 250mm long. If a force of 40N is exerted at its end, find the moment of the force.

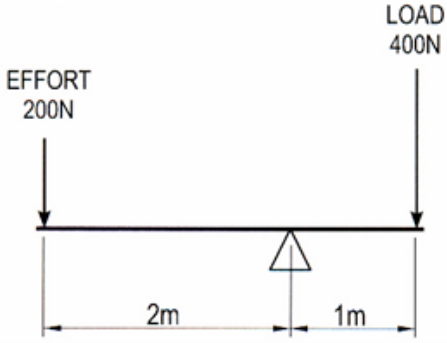


M = F × d

M = 40 × 0.250

M = 10 Nm

The lever system shown opposite is a lever that is in a state of equilibrium (balanced). The input effort is turning the lever Anti-Clockwise (AC); the load is turning the lever Clockwise (C). The forces on each end of the lever are exerting a moment: one clockwise, the other anticlockwise. If the beam (lever) is in equilibrium, **both** of these moments **must be equal**.



Worked example:

Using the lever system above, use the 'principle of moments' to show that the lever is in equilibrium.

For equilibrium to occur:

$$\sum CWM = \sum ACWM$$

$$F_1 \times d_1 = F_2 \times d_2$$

The load is exerting a ClockWise Moment; that is, it is tending to make the lever turn clockwise (CWM).

$$\sum CWM = 400\ N \times 1\ m = 400\ Nm$$

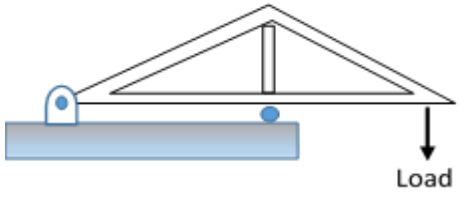
The effort is exerting an Anti-ClockWise Moment (ACWM).

$$\sum ACWM = 200\ N \times 2\ m = 400\ Nm$$

Therefore: $\sum CWM = \sum ACWM$ (Therefore the lever is in a state of equilibrium.)

Moment of a force

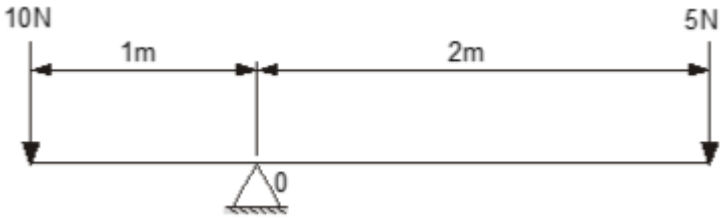
The moment of a force is the turning effect of that force when it acts on a body. The load acting on the frame structure shown will have a turning effect on the structure and cause the support to support different loads.



The Principle of moments states that if a body is in Equilibrium, the sum of the clockwise moments (given positive + sign) is equal to the sum of anti-clockwise moments (negative - sign). This can be written:

$$\Sigma M_O = 0$$

Example 1: Calculate the moments acting below:



$\Sigma M_O = 0$

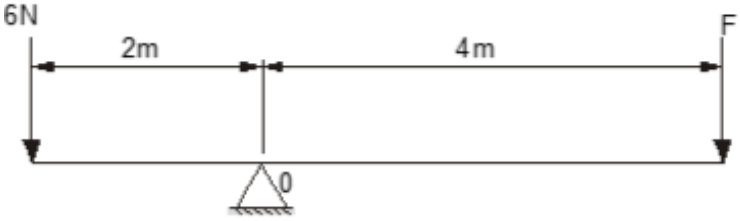
$(5 \times 2) - (10 \times 1) = 0$

$10 - 10 = 0$

The above example is in equilibrium.

We can use this principle to find an unknown force or unknown distance.

Example 2



$\Sigma CWM = \Sigma ACWM$

$(4 \times F) = (6 \times 2)$

$4F = 12$

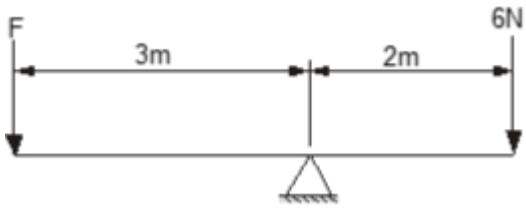
$F = 12 \div 4$

$F = \underline{\underline{3N}}$

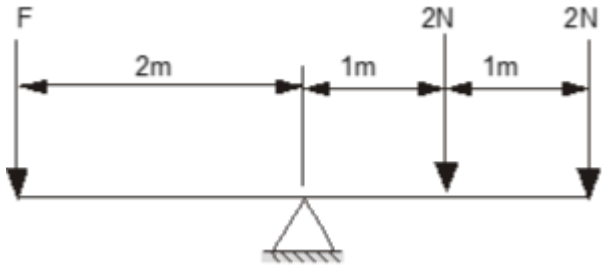
6. Moments

The beams shown below are in equilibrium. Find the unknown quantity for each arrangement.

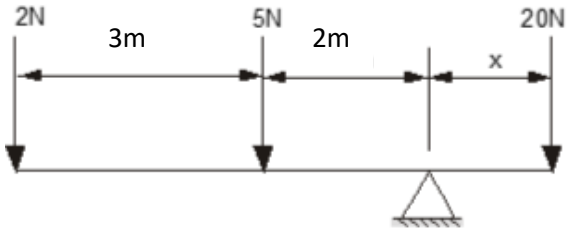
a)



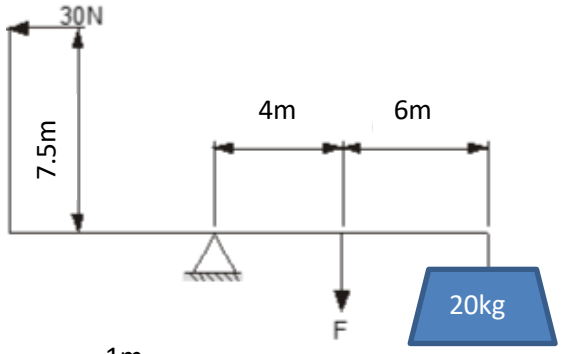
b)



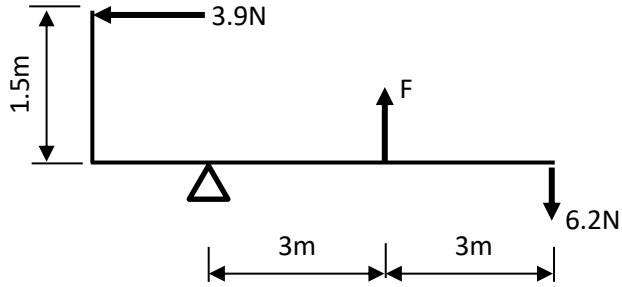
c)



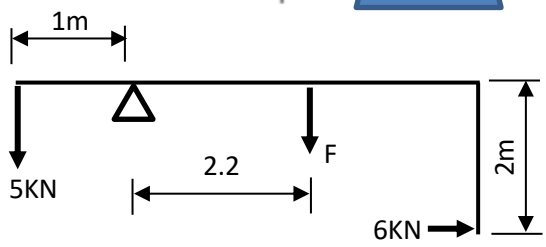
d)



e)



f)



Beam reactions

We can now look at beams with external forces acting on them, resolve moments into their component parts and use moments to find the support reactions. Definitions of some of the terms you have met already:

Resultant: the resultant force is the single force that replaces a system of forces and produces the same effect as the system it replaces.

Equilibrium: This is the word used to mean balanced forces. A body with several forces on it which all balance each other is said to be in a state of equilibrium. A body in equilibrium can be at ease or travelling with uniform motion in a straight line.

Conditions of equilibrium

In the force system in this section you shall apply the **three** condition of equilibrium that you have used before. To solve the force systems the conditions of equilibrium are applied in a certain order, the correct order is shown below.

ΣMo (sum of Moments) = 0

ΣFv (Forces Vertical) = 0

ΣF_H (Forces Horizontal) = 0

Supporting a beam

A beam is usually supported at two points. There are two main ways of supporting a beam.

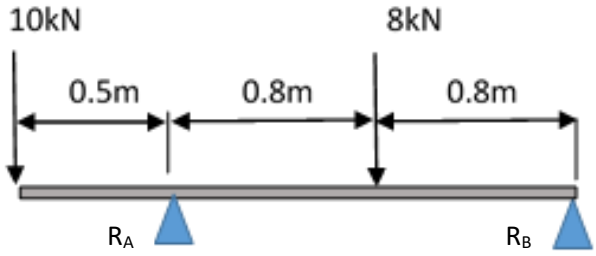
1. Simple supports (knife edge)



Simple support example

These are used when there is no sideways tendency to move the beam. Consider the beam opposite. The forces at the supports, called reactions, always act vertically. The beam is in equilibrium so the conditions of equilibrium apply. The reactions at R_A and R_B are found as follows. Take moments about R_A.

The value of Reactions **R_A** and **R_B** are found as follows. Take moments about R_A



$\Sigma MR_A = 0$

$\Sigma CWM = \Sigma ACWM$

$(8 \times 0.8) = (R_B \times 1.6) + (10 \times 0.5)$

$6.4 = R_B \times 1.6 + 5$

$1.6R_B = 6.4 - 5$

$R_B = \underline{1.4} = \underline{0.88KN}$

1.6

There are two methods available to find R_A

Take moments again, this time about R_B, or equate the vertical forces (since the beam is in equilibrium)

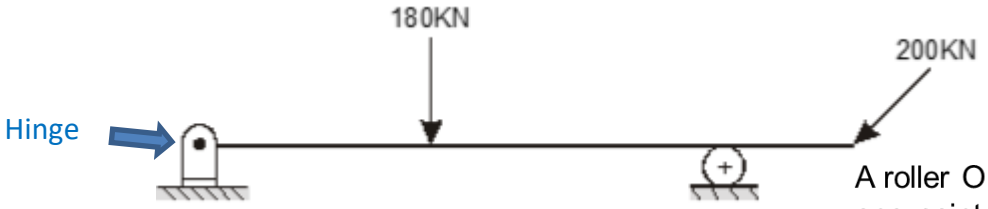
$(R_A + R_B) = 10 + 8$

$R_A = 18 - R_B$

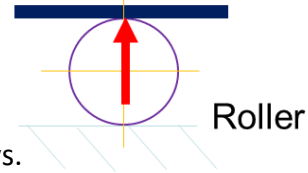
$R_A = 18 - 0.88$

$R_A = \underline{17.12KN}$

2. Hinge and roller



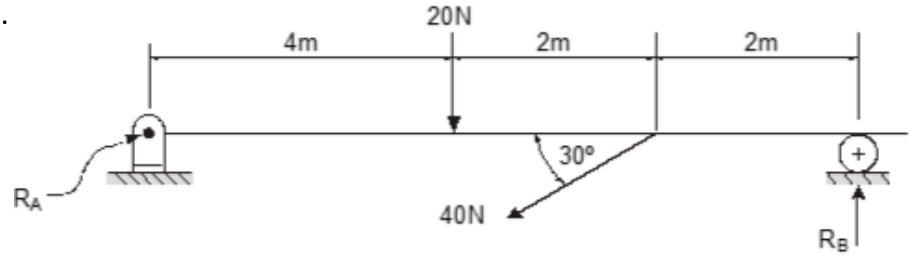
A roller ONLY touches the beam at one point. It is a vertical force acting the centre of the roller.



Hinge and roller support example

These are used when there is a possibility that the beam may move sideways.

Note: The reaction at a roller support, R_B, is always at right angles to the surface. The direction of R_A is assumed. If any of the components work out as negative values, then the direction will be opposite the assumed direction.



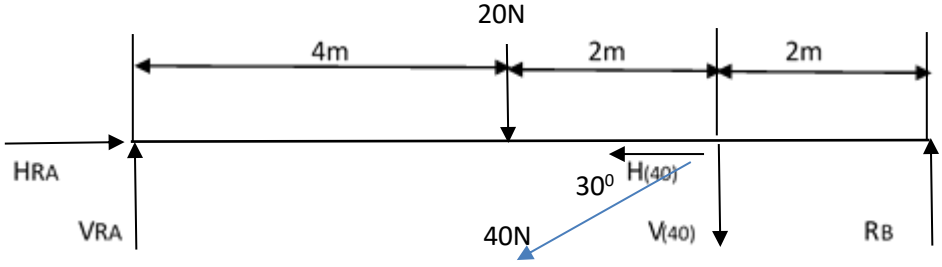
The reaction at the hinge support can be any direction. (Find the two components of the hinge reaction, then the resultant and angle).

There are three unknown quantities above: -

1. The magnitude of Reaction **R_B**.
2. The magnitude of Reaction **R_A**.
3. The direction of Reaction **R_A**.

Beam reactions: worked examples

A free-body diagram showing vertical and horizontal components of the forces is shown below.



The horizontal and vertical components of the angled 40N force are found first.

$F_H = F \times \cos°$

$F_H = 40 \cos 30° = 34.64N$

$F_V = F \times \sin°$

$F_V = 40 \sin 30° = 20N$

To find R_B, take moments around V_{RA}. This eliminates one of the unknown forces. Then calculate vertical and horizontal forces.

$\Sigma M_A = 0$

ACWM = CWM

$R_B \times 8 = (+20N \times 4m) + (40Sin30° \times 6m)$

$8R_B = 200$

$R_B = \frac{200}{8} = 25N$

Forces Vertical

$\Sigma Fv = 0$

$\Sigma Fv \uparrow = \Sigma Fv \downarrow$

$R_A + R_B = 20N + 40Sin30°$

$F_V R_A = 40 - R_B$

$F_V R_A = 40 - 25 = 15N$

Forces Horizontal forces

$\Sigma F_H = 0$

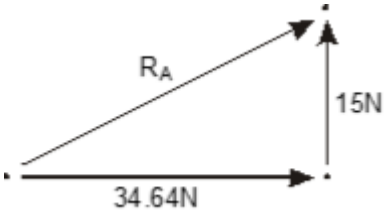
$\Sigma F_H \rightarrow = \Sigma F_H \leftarrow$

$\Sigma F_H R_A - 40Cos30°$

$\Sigma F_H R_A - 34.64 = 0$

$F_H R_A = 34.64N$

1. Use Force Vertical (F_V) and Force Horizontal (F_H) to find Resultant R_A.

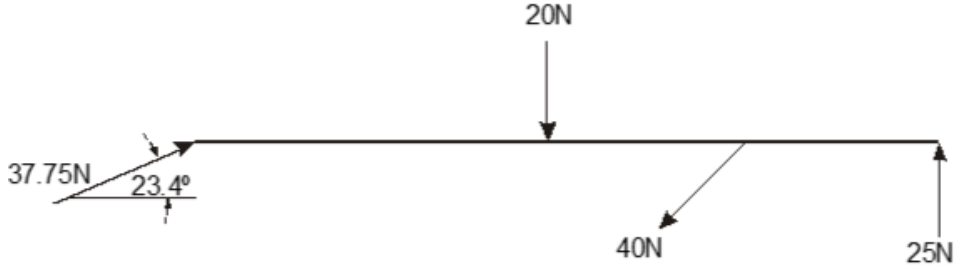


$$R_A = \sqrt{15^2 + 34.64^2}$$
$$= \sqrt{225 + 1199.93}$$
$$= 37.75N$$

2. Find direction of R_A

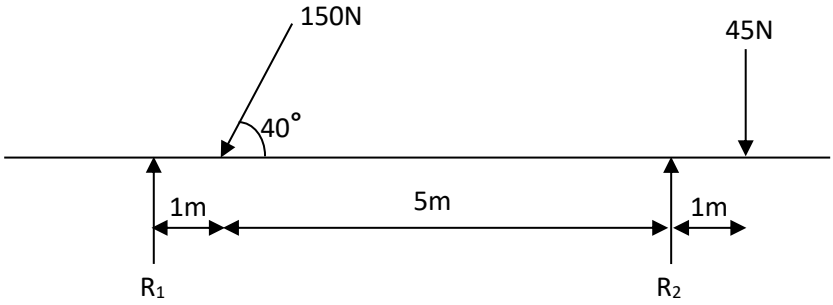
$$\tan a = \frac{VR_A}{HR_A}$$
$$= \frac{15}{34.64}$$
$$= 23.4^\circ$$

Results shown in a free body diagram below:



Worked example

When there are two unknown forces: R_1 and R_2



Step 1: You need to take moments around one of the unknown forces:

$\Sigma MR_1 = 0$

ACWM = CWM

$6m \times R_2 = (Fv \times angle \times dist) + (45N \times 7m)$

$6m \times R_2 = (150N \times \sin 40° \times 1m) + (45N \times 7m)$

$6R_2 = 96.42 + 315$

$R_2 = 411.42 \div 6$

$R_2 = 68.57N$

First: Make R1 a pivot point to take moment around.

Second: What forces are making a downward but also ClockWise Moment? (Fv) 150NSin40 and 45N

Step 2: Now you can use $\Sigma F_V = 0$:

Forces up = Forces down

$R_1 + R_2 = (150 \sin 40°) + 45N$

$R_1 = (150 \sin 40°) + 45N - 68.57$

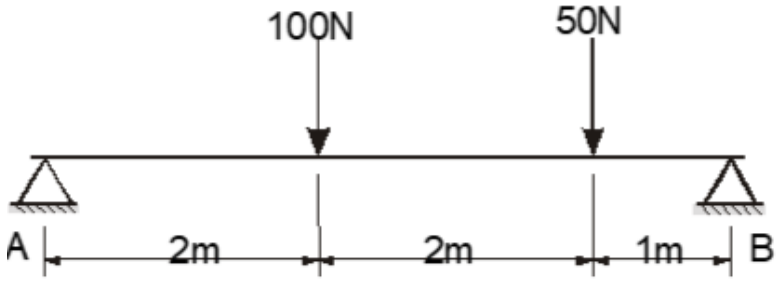
$R_1 = 72.85N$

Note opposite: The distance from the point of rotation to where the angled Force is coming in contact with the beam is NOT required when calculating the Vertical force (150Sin40°).

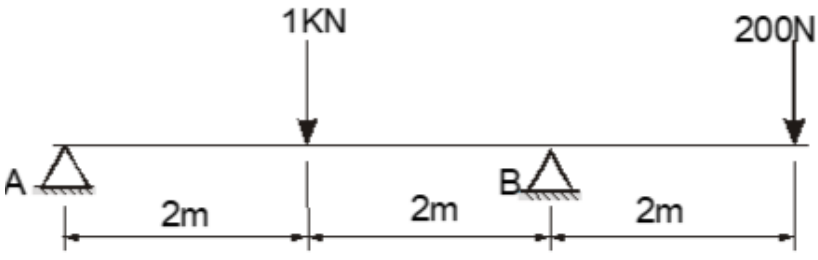
Assignments: Beam reaction

7. Find the reactions at supports A and B for each of the loaded beams shown below.

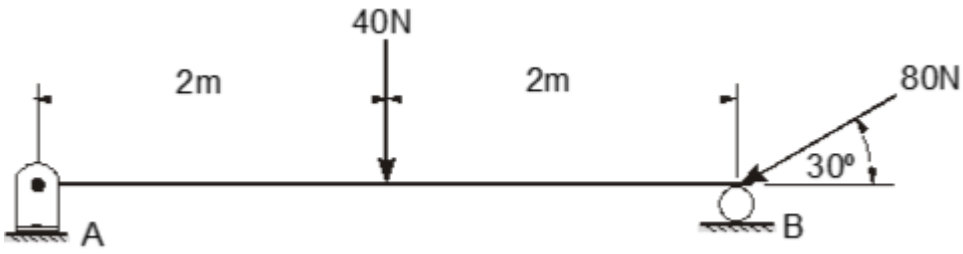
a)



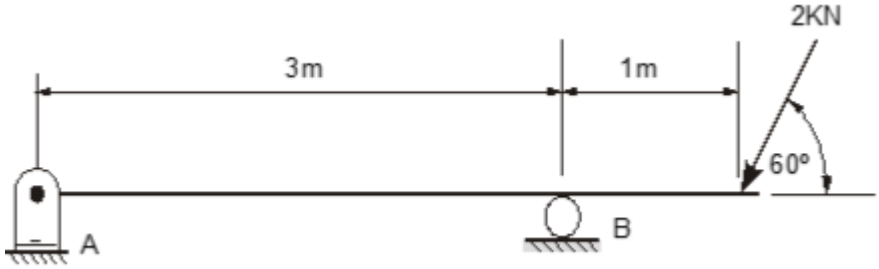
b)



c)



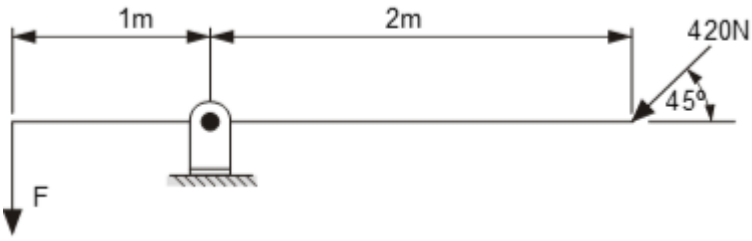
d)



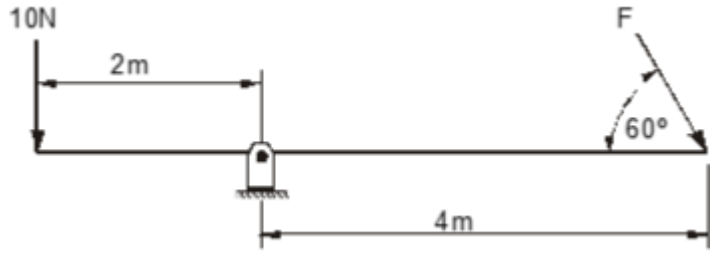
8.

The following beams, in equilibrium, have inclined forces. Calculate the unknown quantity.

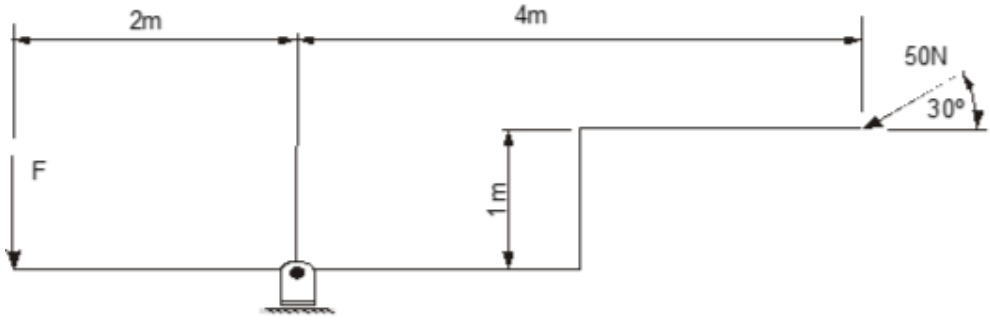
a)



b)



c)

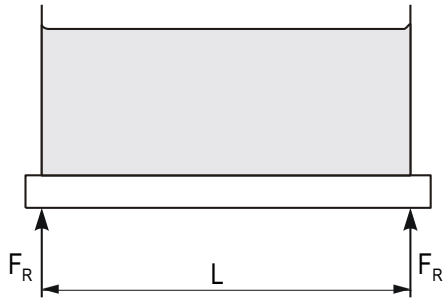


Uniformly Distributed

Load (UDL) Beam reactions

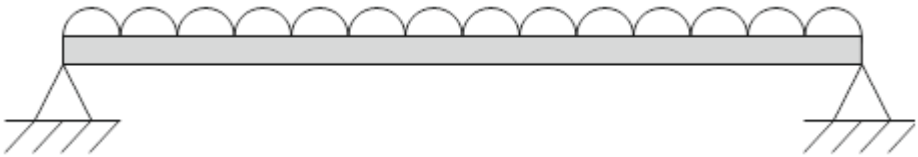
In many structures, a UDL is a load which is spread continuously and evenly along the whole length of a beam, rather than at a single point. For example, the supporting steel beam in a building will be supporting a floor. In practice, this is the usual type of load a beam will require to support due to its self weight and often due to the way it is loaded.





'ω' is the load acting along the length of the beam and is given in terms of the total force acting on each metre length of the beam W = **ω** L and the force is measured in Newtons per meter (ω = N/m).

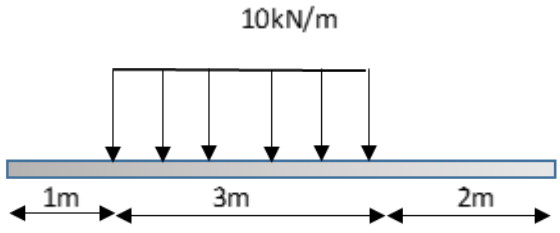
The floor will exert a force over the entire length of the beam rather than at one particular point. We can represent this type of force in the method shown below.



When calculating moments, it is important to calculate the total force applied across the beam. This total force is then calculated using the centre of gravity (almost always the middle) of the beam. There will be occasions where single point loads must also be considered in calculations.

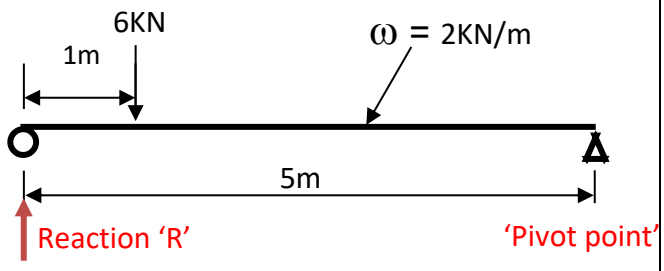
The symbol to represent the UDL of a beam is ω.

The magnitude of the resultant force is equivalent to the area under the curve of the distributed load multiplied by the load per metre. In the example opposite the resultant would be 10kN/m x 3m = 30kN



Worked example

Q) Find the reaction force at the roller ball.



$\Sigma M_{pivot} = 0$

CWM = ACWM

(6000N x 4m) + (2000N x 5m x 2.5m) = F_R x 5

Important note: 2KN in one meter (2KN/m). Beam is 5 meters long so 2KN x 5m. However, it must also include the centre of gravity of the beam in the middle of the beam which is the UDL distance divided by 2. This is 2.5m. Therefore the UDL at midpoint is: 2000N x 5m x 2.5m = 25 000N

6000N x 4m = 24 000N

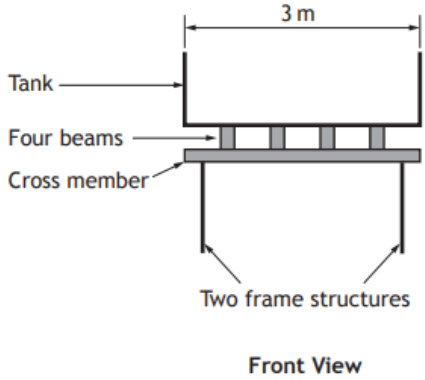
24 000 + 25 000 = $5F_R$

$F_R = \frac{49\,000}{5}$

F_R = 9.8KN

Worked example 2. SQA past paper 2015 Q8

A water storage tank, dimension 3m x 3m, sits on 4 timber beams. The beams are secured to a cross member at one end and fixed to a wall at the other. Each beam carries an equal share of the tanks weight.



Front View

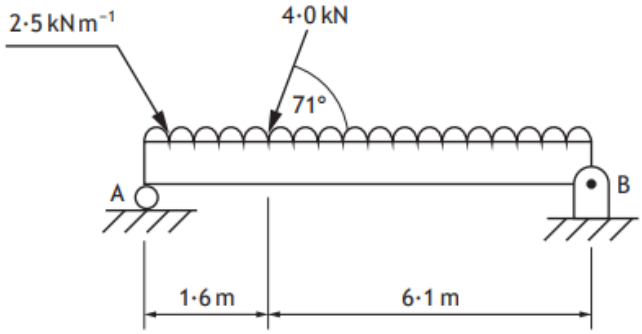
a) Calculate the maximum uniformly distributed load exerted on each beam if the water level in the storage tank does not exceed 1.5 metres. Ignore the weight of the tank. ($1m^3$ of water has a mass of 1000 kg.)

Answer

Expected Answer(s)	Max Mark	Additional Guidance
Volume of tank = l × b × h = 3 × 3 × 1·5 = 13·5m^3 Therefore mass = 13·5m^3× 1000kg = 13500Kg **(1)** Therefore Force = mass × gravity = 13500kg × 9·8 = 132300 N (132·3 KN) **(1)** If there are four beams the total force will be divided by four 132300 ÷ 4 (Beams) = **33075 N** **or** 33·08 KN **(1)** If the load is uniformly distributed across **3** metres therefore 33075 ÷ 3m = 11025 Nm^{-1} **11025 Nm^{-1}** or 11·03kNm^{-1} answer and unit **(1)**	**4**	Units are not required for the mark for intermediate steps Accept 11036Nm^{-1} if they use 9·81 for gravity

Worked example SQA past paper 2017 Q3

A beam to be used in the construction of a new holiday cottage is being tested.

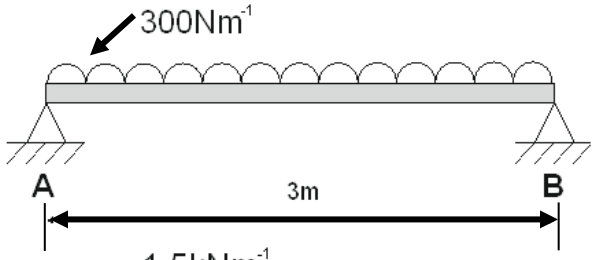


Calculate, by taking moments about B, the vertical reaction at A.

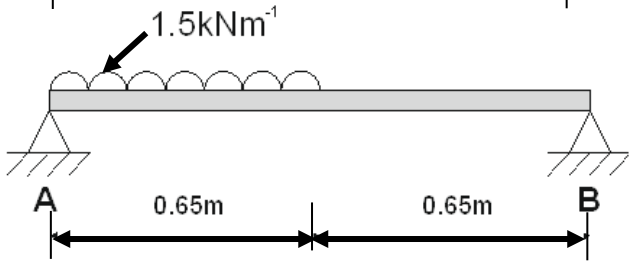
Expected answer(s)	Max mark	Additional guidance
Uniformly distributed load: 2.5 x 7.7 = 19.25 kN @ 3.85m midpoint	3	1 mark for value 19.25 kN
Moments about B: (6.1 x 4.0 sin71) + (19.25 x 3.85) = 7.7 R_A 23.07 + 74.11 = 7.7 R_A		1 mark for substitution
R_A = 97.18/7.7 = 12.6 kN		1 mark for final answer with unit

9. Calculate the reaction forces in the following UDL examples.

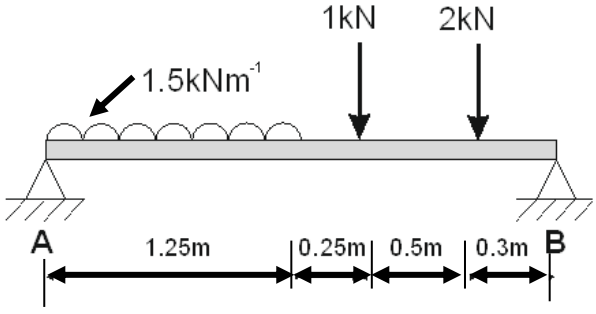
a)



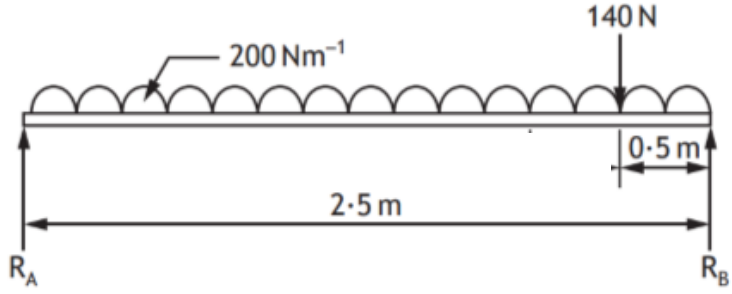
b)



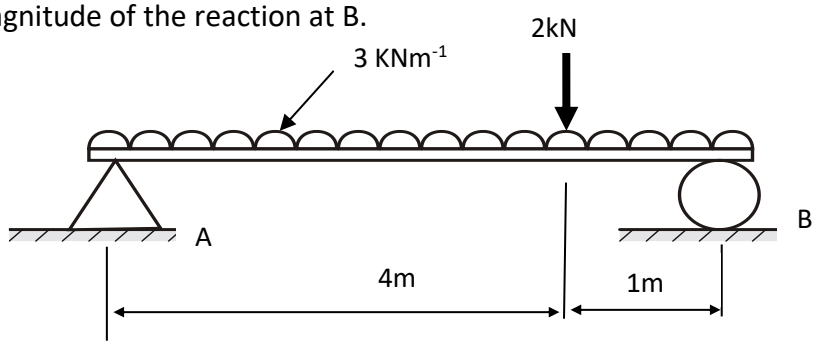
c)



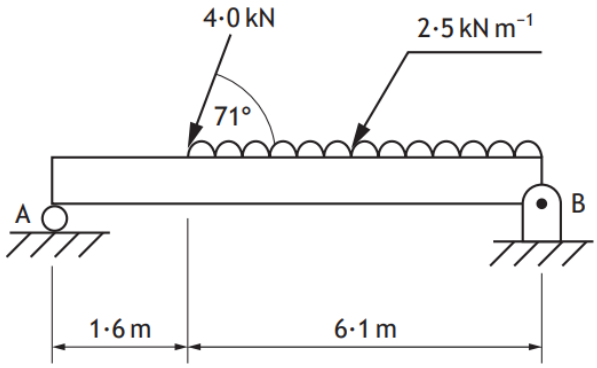
d) Calculate the reaction force, R_B.



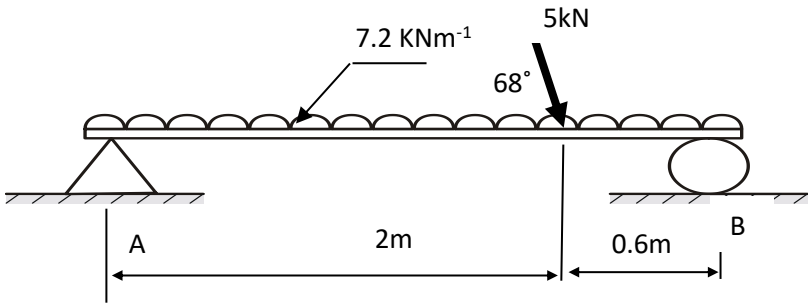
e) Calculate the magnitude of the reaction at B.



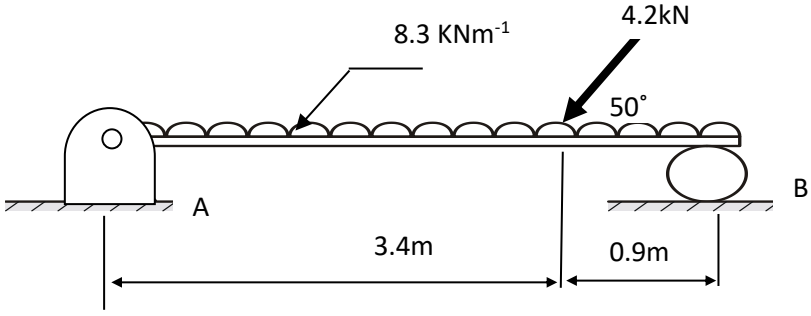
f) Calculate both reaction forces in the following UDL examples.



g)



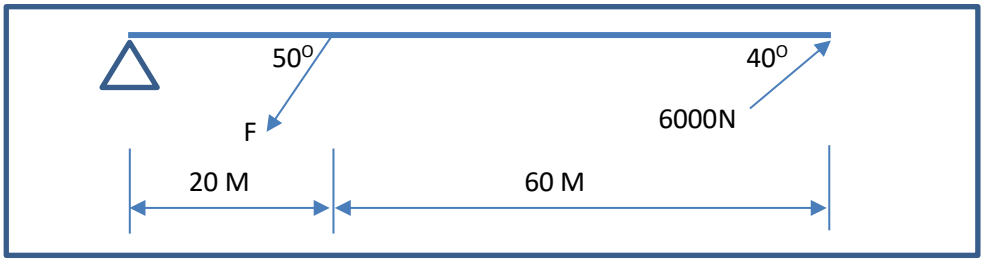
h)



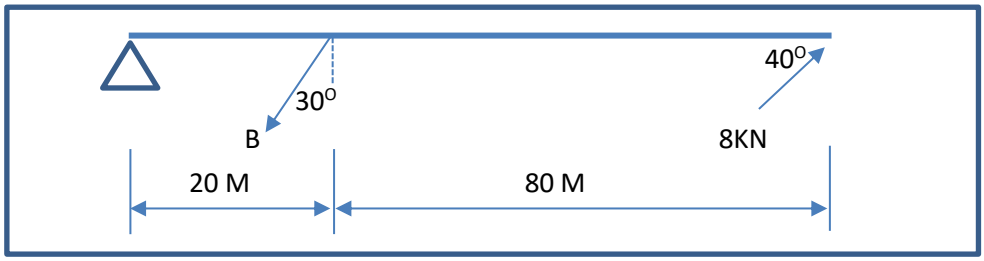
10.

Additional beam questions.

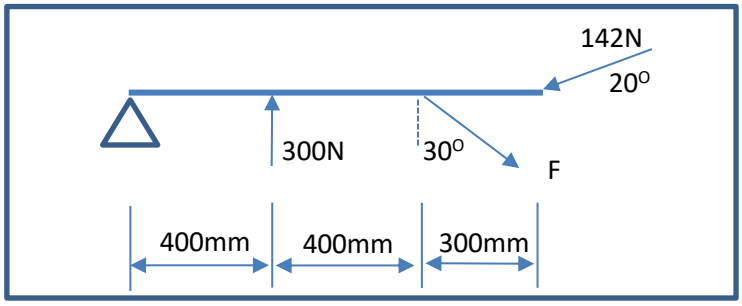
A. Calculate the unknown Force F.



B. Calculate the unknown Force B.

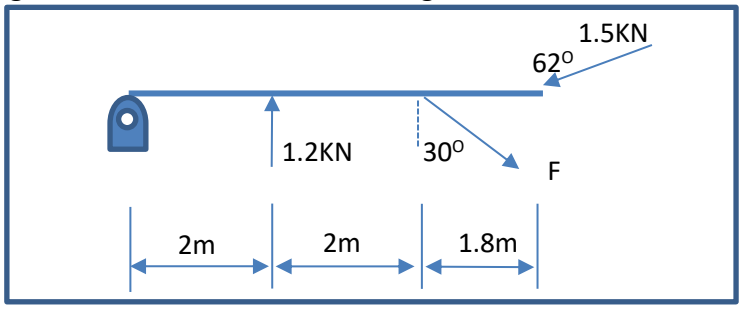


C. Calculate the magnitude of force F.



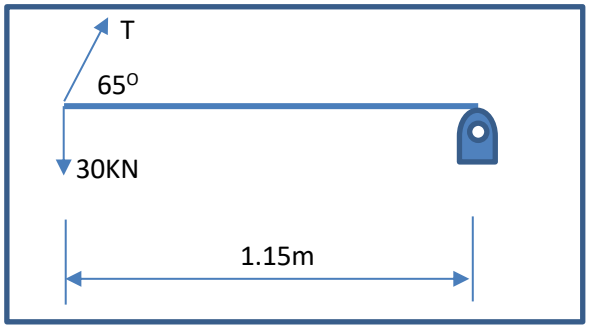
D. (i) Calculate the magnitude of Force F.

(ii) Calculate the magnitude and direction at the hinge.



E. A cable (T) is used to support a shop sign attached to a wall, calculate the magnitude of Force T.

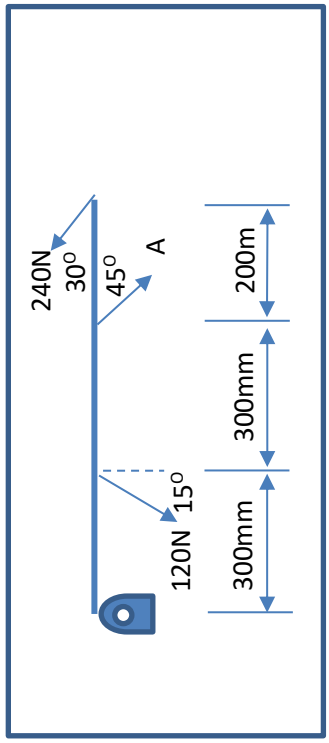
(i) Calculate the magnitude and direction of the reaction at the hinge.



F. A tv is attached to the wall using a wall bracket. The image shown on the right displays the forces acting on the bracket.

(i) Calculate the force, A, exerted by the wall anchor.

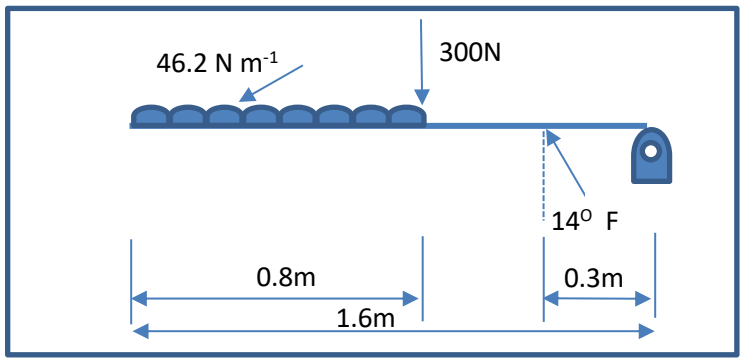
(ii) Calculate the magnitude and direction of the reaction force at the hinge.



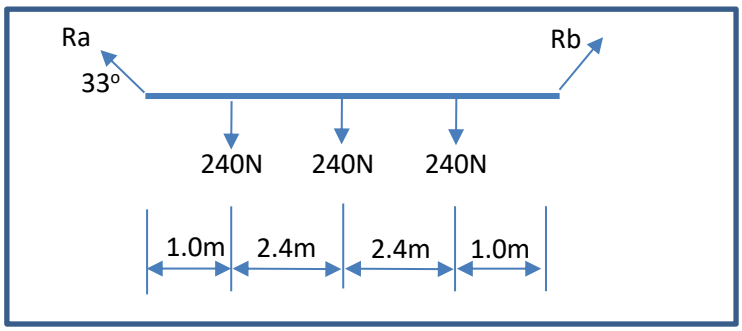
G. A uniformly distributed load is 46.2KN m^{-1}

(i) Calculate the magnitude of force F. (3)

(ii) Calculate the magnitude and direction at the hinge. (4)

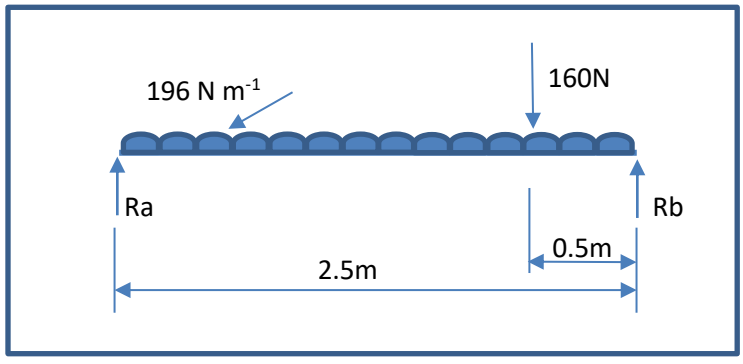


H. Calculate, by taking the moments, the reactions force Ra (3)

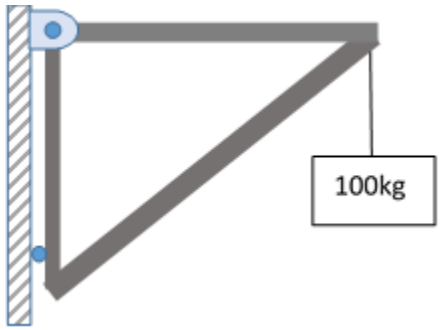


I. Flat pack furniture and other items are stored on a shelving in a home improvement store. The shelving must be design ed to support both point loads and uniform distributed loads. A simplified free body diagram of the forces expected on one shelf is shown below.

(i) Calculate the reaction force Rb. (3)



Framed structures

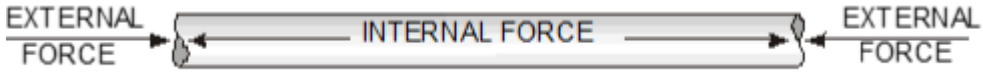


A frame structure is an assembly of members and joints (usually called nodes) which is designed to support a load. Examples of frame structures include roof trusses, bridges, pylon towers etc.

The members in this framed structure can be as 'ties' or 'struts', depending on the type of force they support.

Strut

Members that are in **compression** (crushed), due to external forces trying to compress them, are known as struts.



Tie

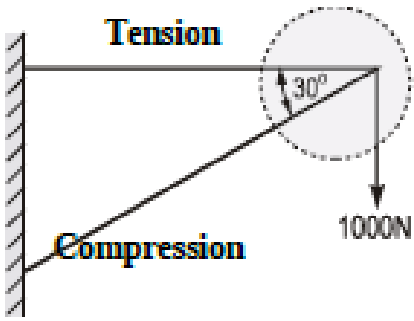
Members that are in **tension** (stretched), due to external forces trying to pull them apart, are known as ties.

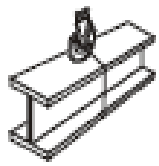


Example

The small crane, shown, is used in a steelworks yard to lift 'I' beams of weight 1000N. Find the forces in the compression and tension members.

Scale 10mm = 1m.

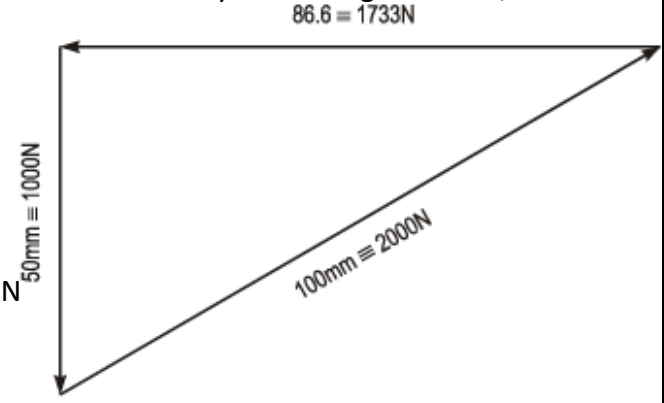




To find the forces created in the **tension** and **compression** members by the 1000 N weight (load), draw the triangle from the area circled. Select a suitable scale and then draw the known force first, the 1000 N load. A line is drawn through one end of the load line parallel to one of the unknown forces. Another line is drawn through the other end, parallel to the second unknown force. By measuring each line, the size of each force can be found.

(Note: the arrowheads must follow round the triangle.)

Vector or Force diagram



Scale: 10mm = 200 N

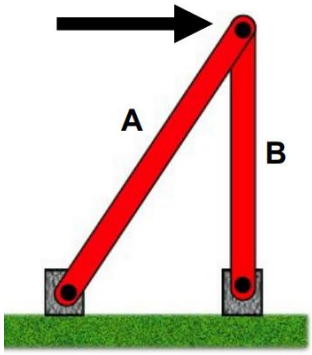
By scale measurement, the compression member = 2000 N

By scale measurement, the tension member = 1733 N

Framed structures

The Frame Structures below are all under load in the direction of the arrows. Sketch the frames into your jotter and label whether each member is under compression or tension and if each member is a strut or a tie. Remember a TIE is under Tension (being pulled apart) and a STRUT is under compression (being squashed).

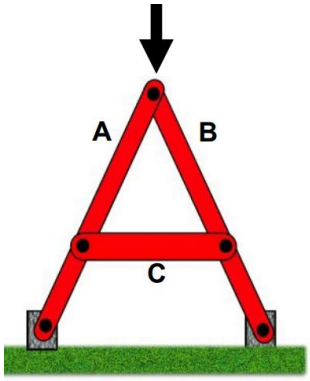
Frame one



Member A is under _____ and this means it is a _____.

Member B is under _____ and this means it is a _____.

Frame two

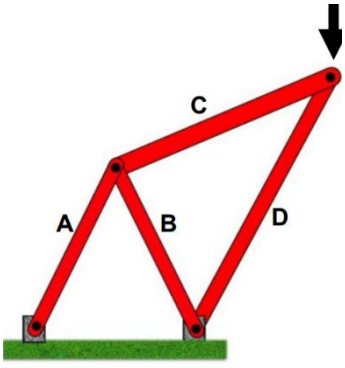


Member A is under _____ and this means it is a _____.

Member B is under _____ and this means it is a _____.

Member C is under _____ and this means it is a _____.

Frame three



Member A is under _____ and this means it is a _____.

Member B is under _____ and this means it is a _____.

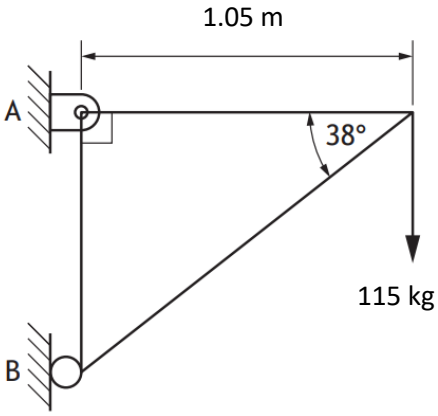
Member C is under _____ and this means it is a _____.

Member D is under _____ and this means it is a _____.

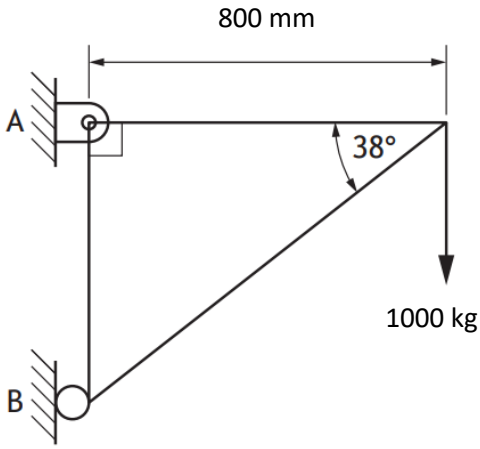
Frame reactions

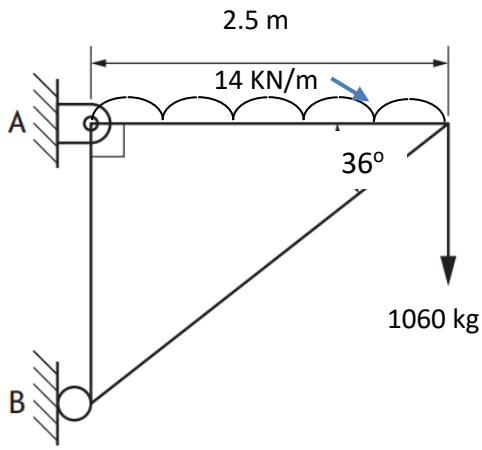
Worked example.

A shop signage frame structure is shown below. Calculate the magnitude and indicate the direction of the reaction at B.



11. Calculate the reactions at supports A and B for each of the loaded frames shown below.





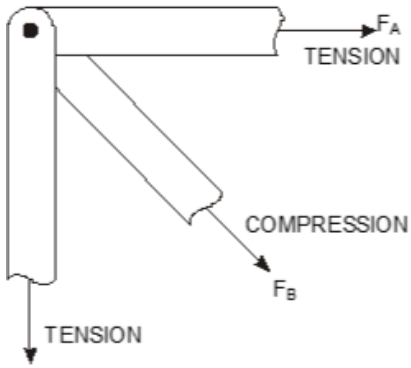
Nodal analysis

There are several methods of solving frame structure problems. The method we shall use is called Nodal Analysis. Any joint where members meet is known as a Node.

This method relies on the fact that if structures are in equilibrium then each node will be in equilibrium. The sum of the forces acting on any the node will zero.

Nodes

The members are either in compression (strut) or tension (tie). They can be represented at the node as shown opposite.



F_B is under compression and pushes into the node, **F_A** is under tension and pulls away from the node.

Conditions of static equilibrium

SUM OF THE MOMENTS = 0	$\Sigma M_o = 0$
SUM OF THE FORCES VERTICAL = 0	$\Sigma F_V = 0$
SUM OF THE FORCES HORIZONTAL = 0	$\Sigma F_H = 0$

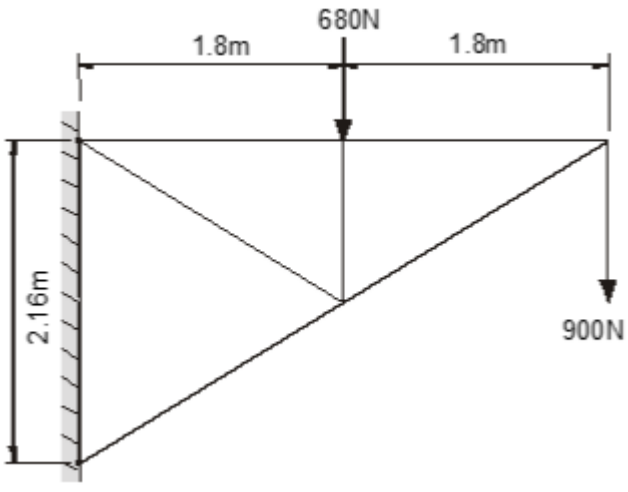
When solving frame structures some analysis of the structure is required to determine the best starting point and which of the conditions of static equilibrium to apply first.

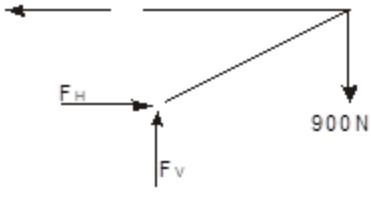
Nodal analysis: Solving simple frame structures

To help solve frame structure problems there are some simple rules to follow depending on the type of structure.

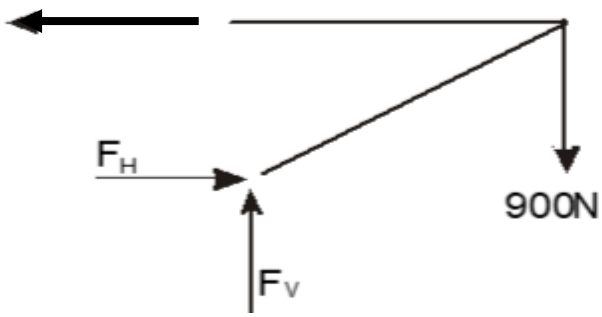
Cantilevered frame structures

For this frame structure it is not necessary to use moments to help find the forces in the members. The node with the 900N load acting has only one unknown vertical component.



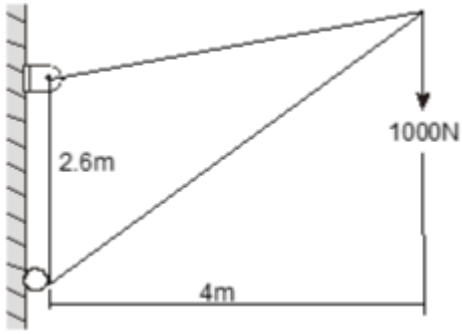


As all vertical forces acting on the node must equal zero then $\mathbf{F_V}$ must equal 900N. By using trigonometry



it is now possible to find the other forces acting on this node. In this example all nodes have more than one unknown force or component of a force.

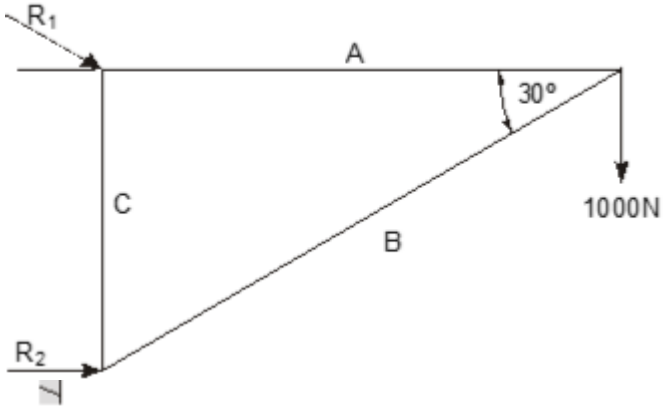
In the example below however, all nodes have more than one unknown force or component of a force. To solve this frame structure, take moments about the top support to find the reaction force at the roller support.



A free body diagram of the frame structure is shown below.

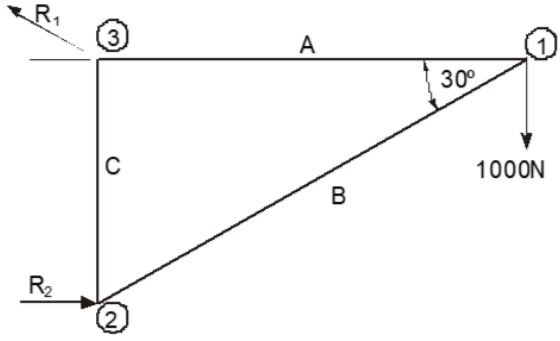
The framework is supported at two points. The hinge support at the top is being pulled away from the wall. R1 will act against this pull and keep the hinge attached to the wall. The roller support is being pushed into the wall. R2 will act against this force and in the opposite direction as in the diagram. As member B is acting on a roller then R2 will be at 90° to member C.

As $\mathbf{R_1}$ is acting at a hinge support there will be a vertical and horizontal component. At this stage guess the direction of $\mathbf{R_1}$.



If a wrong assumption about the member being in tension or compression or the direction of a force is made then a negative value will be produced from the calculation. If this is the case the direction of the force is simply reversed.

To find the forces in the members and the reactions at the supports, study each of the structure nodes separately. Each node is in equilibrium so we can use the conditions of equilibrium.



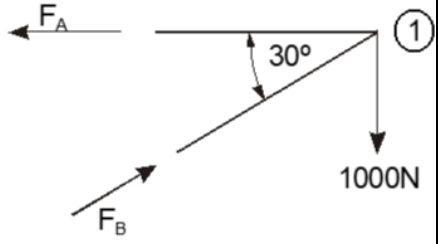
Sum of the Moments = 0 ΣMo = 0

Sum of the Forces vertical = 0 ΣFV = 0

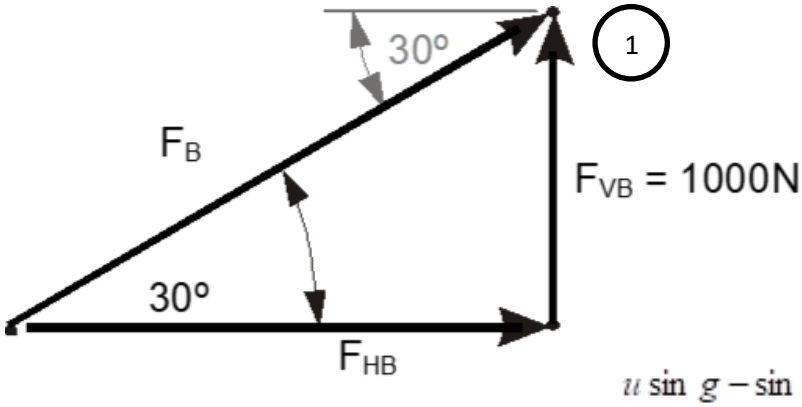
Sum of the Forces horizontal = 0 ΣFH = 0

In this sort of problem where the structure is cantilevered out from the wall, it is normally solvable without considering the moments. Start by finding a node that has only one unknown vertical or horizontal component.

Calculating node 1



The forces acting on node 1 are shown opposite.
Note an assumption has been made about the direction of F_A and F_B. F_A is assumed to be in tension and acting away from node 1. F_B is assumed to be in compression and acting towards node 1.
As the sum of Vertical forces is equal to zero then the vertical component of **F_B** (**F_{VB}**) is 1000N acting up. From this we can find **F_B** - we can redraw **F_B** and the two components **F_{HB}** and **F_{VB}** to form a triangle as shown below.



$$u \sin g - \sin \theta = \frac{opp}{hyp}$$

The force in member **A** (**F_A**) will equal and opposite to **F_{HB}**.

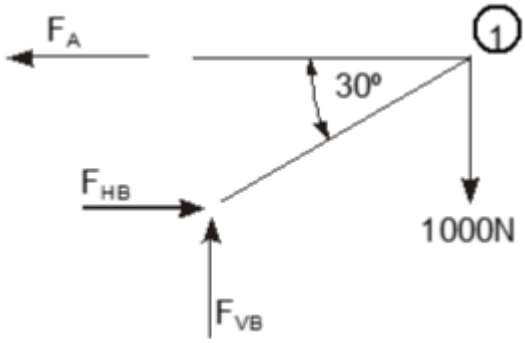
$$\sin 30° = \frac{F_{VB}}{F_B}$$

$$F_B = \frac{F_{VB}}{Sin30°}$$

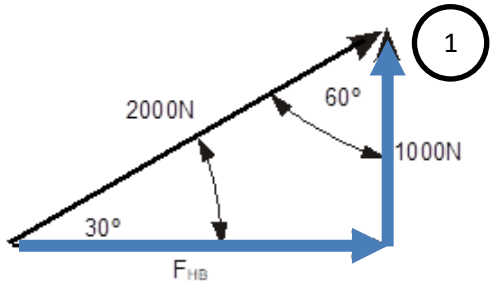
$$= \frac{1000}{0.5}$$

$$= 2000N$$

Find **F_{HB}** -



This diagram shows all the forces acting at node 1.



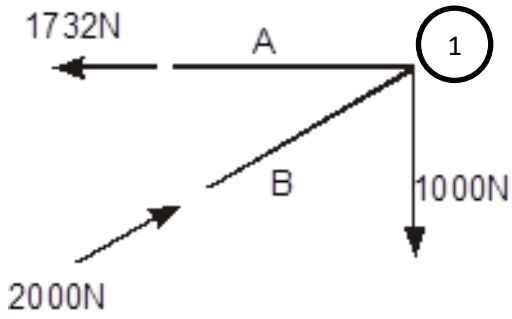
$$\cos 30° = \frac{F_{HB}}{F_B}$$

$$F_{HB} = \cos 30° \times F_B$$

$$= 0.87 \times 2000$$

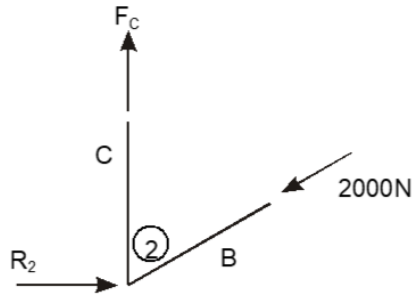
$$= 1732N$$

From above **F_A** = 1732N

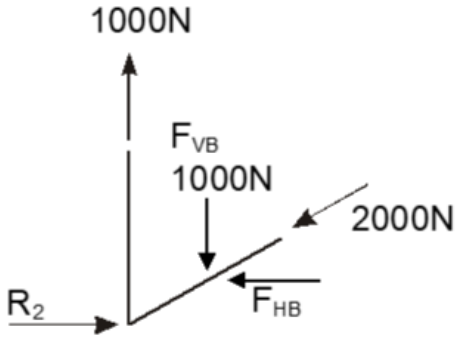


Calculate node 2

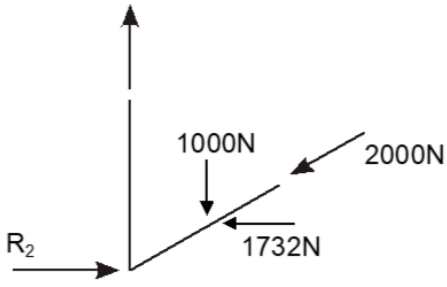
There are three forces acting at node 2.



During analysis of node 1, $\mathbf{F_B}$ was found to be 2000N and was not negative so the assumed direction towards the node (compression) was correct. In node 1, $\mathbf{F_B}$ is shown to be acting as a compressive force towards the node with a magnitude of 2000N.

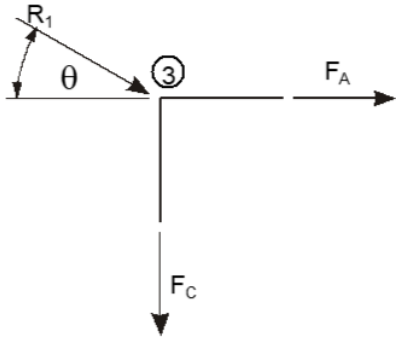


$\mathbf{F_C}$ will be equal and opposite to the Vertical component of $\mathbf{F_B}$ ($\mathbf{F_{VB}}$). From analysis of node 1 $\mathbf{F_{VB}}$ was equal to 1000N therefore: $\mathbf{F_C}$ = 1000N. As $\mathbf{R_2}$ is a roller support the reaction will be at 90° to the surface. $\mathbf{R_2}$ will be equal to the Horizontal component of $\mathbf{F_B}$. From above $\mathbf{F_{HB}}$ was equal to 1732 N therefore: $\mathbf{R_2}$ = 1732N
The forces acting at node 2 are shown below.



Calculate node 3

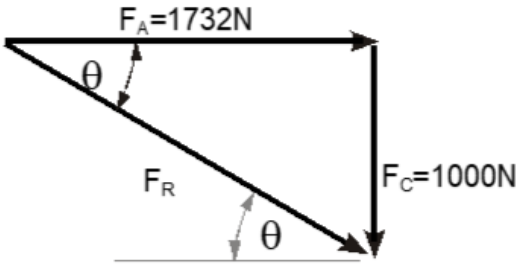
Node 3 has three forces acting on it.



R_1 will be equal and opposite to the resultant of **F_A** and **F_C**.
From analysis of node 1 **F_A** was found to be 1732N.
From analysis of node 2 **F_C** was found to be 1000N
Find the resultant using Pythagoras.

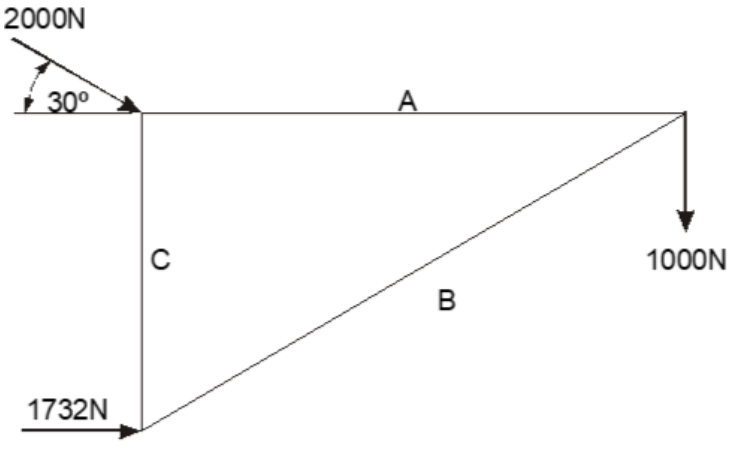
$$F_R = \sqrt{F_A{}^2 + F_C{}^2}$$
$$= \sqrt{1732^2 + 1000^2}$$
$$= 2000N$$

Find angle of resultant (**θ**) –



$$\tan\theta = \frac{F_C}{F_A}$$
$$= \frac{1000}{1732}$$
$$= 0.58$$
$$\theta = 30^\circ$$

The task was to find the forces in the members and the reactions forces at the supports. The results can be shown as below -



MEMBER	FORCE	TYPE OF FORCE
A	1732N	TENSION
B	2000N	COMPRESSION
C	1000N	TENSION

Worked example

A frame structure is shown in Figure Q6.

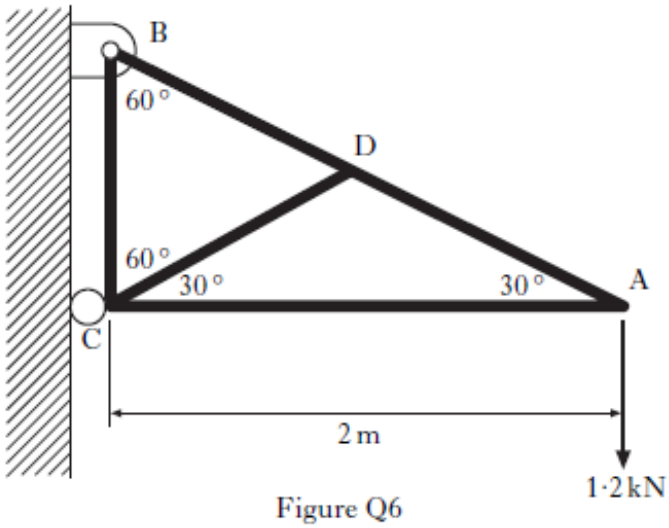


Figure Q6

(*a*) Calculate the **magnitude** of:

(i) the reaction force at C, using the Principle of Moments;

(ii) the reaction force at B.

CD is a redundant member.

(*b*) Explain what is meant by the term "redundant member".

(*c*) Calculate, using *nodal analysis*, the **magnitude** and **nature** of the forces in members AD and AC.

a)

$$BC = \tan 30 \times 2$$
$$= 1{\cdot}15\text{m}$$

i)

$$\Sigma M_B = 0$$
$$R_C \times 1{\cdot}15 = 1{\cdot}2 \times 2$$
$$R_C = 2{\cdot}4/1{\cdot}15$$
$$R_C = \mathbf{2{\cdot}08\ kN}$$

ii)

$$R_B = \sqrt{(1{\cdot}2^2 + 2{\cdot}09^2)}$$
$$R_B = \mathbf{2{\cdot}40\ kN}$$

b) No forces in member

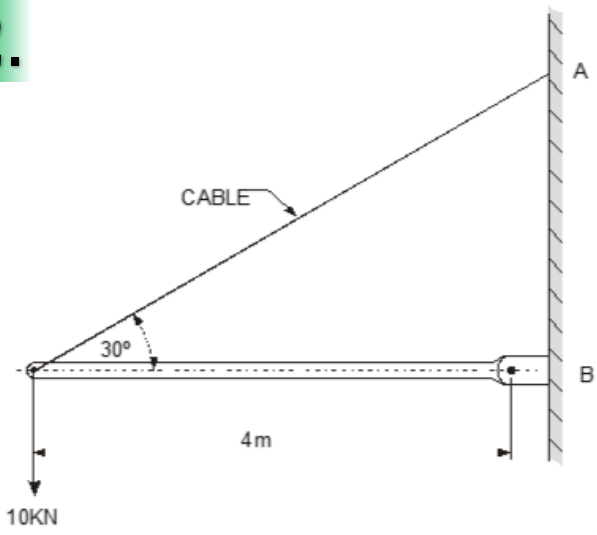
c) Analysing Node A

$$\Sigma F_V = 0$$
$$F_{AD} = 1{\cdot}2/\cos 60$$
$$= \mathbf{2{\cdot}40\ kN}\quad \mathbf{Tie}$$

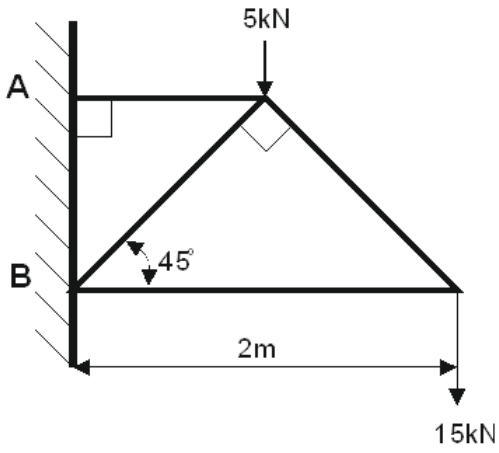
$$\Sigma F_H = 0$$
$$F_{AC} = 2{\cdot}4 \times \cos 30$$
$$= \mathbf{2{\cdot}08\ kN}\quad \mathbf{Strut}$$

Hinge only beams

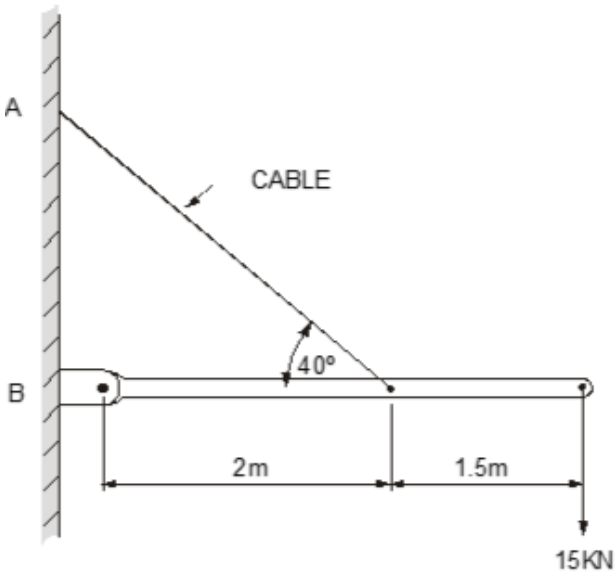
12. Calculate the reactions at support B for each of the hinge only beams shown below.



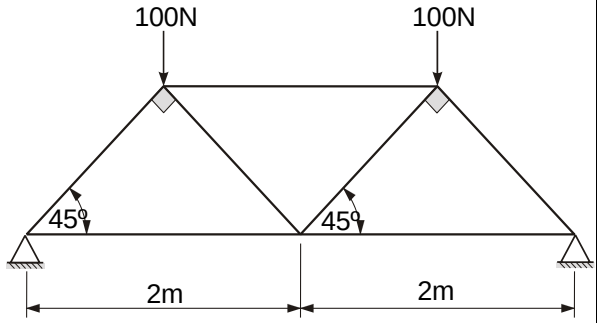
b)



c)



Truss frame structures

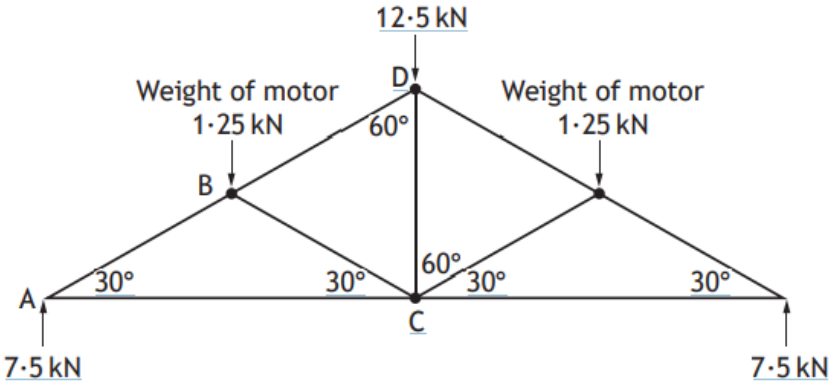


Treat structure like this as a beam and use moments to find reactions. Until the reaction forces are found all nodes have more than one unknown force or component.

Then start analysis at **R_1** or **R_2**, which will now have only one unknown component.

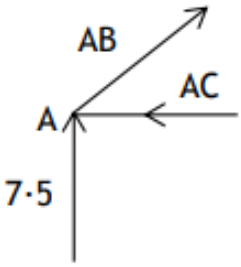
Worked example SQA past paper 2016 Q9

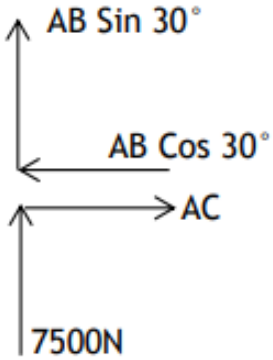
The free body diagram for a pin-jointed frame is shown below for the end support of the roof. Member BC is a 1·1 KN strut. Calculate, using nodal analysis, the magnitude and nature of the force in beams AB, AC, BD. Show all working and final units.



Worked answer on Powerpoint

Node A





$\Sigma F_v = 0$
$AB \text{ Sin } 30^\circ + 7500 = 0$
$AB \text{ Sin } 30^\circ = -7500$
$AB = \frac{-7500}{\text{Sin } 30^\circ}$
$AB = \frac{7500}{0{\cdot}5}$
$AB = 15\text{kN (Strut)}$

$\Sigma F_M = 0$
$AB \text{ Cos } 30^\circ = AC$
$15000 \times 0{\cdot}866 = AC$
$12990{\cdot}38\text{N} = AC$

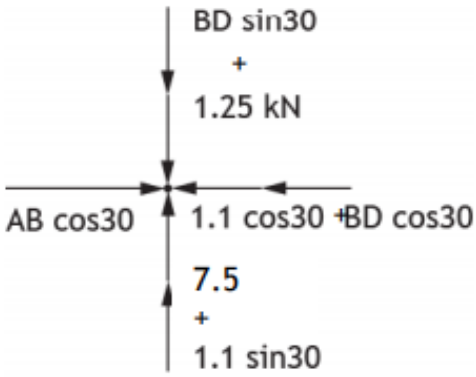
$AC = (12{\cdot}9\text{kN})$ TIE

1 mark for magnitude AB

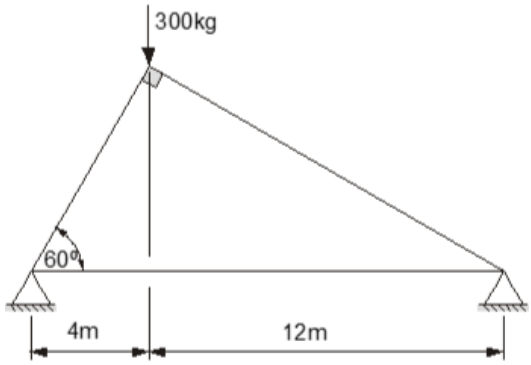
1 mark for nature AB

1 mark for magnitude AC

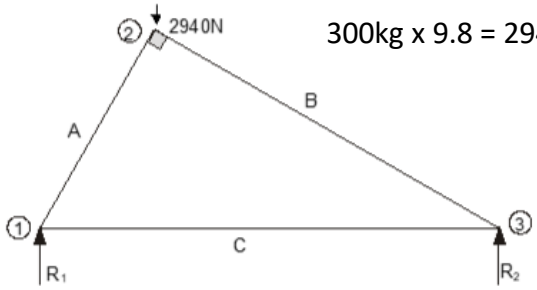
1 mark for nature AC

Expected Answer(s)	Max Mark	Additional Guidance
Node B  $\Sigma F_V = 0$ $1{\cdot}25 + 0{\cdot}5BD = 7{\cdot}5 + 0{\cdot}55$ $BD = \frac{8{\cdot}05 - 1{\cdot}25}{0{\cdot}5}$ BD = 13·6kN Strut		**Either solution for Node B acceptable** **1 mark** for vertical BD (BD sin30) **1 mark** for correct substitution of F_v **1 mark** magnitude of BD **1 mark** for nature of BD
Alternative way of calculating BD below $\Sigma F_H = 0$ AB Cos 30 = 1·1 Cos 30 - BD Cos 30 $BD = \frac{11{\cdot}75}{6{\cdot}866} = 13{\cdot}6\text{kN}$ Strut		**1 mark** for horizontal BD (BD cos30) **1 mark** for correct substitution of F_H **1 mark** magnitude of BD **1 mark** for nature of BD
AC = 12·9kN (Tie) BC = 1·1kN (Strut) BD = 13·6kN (Strut) AB = 15kN (Strut)		

13. a) Calculate the reactions R_1 and R_2 and the forces for the members in the frame structure shown below.



b) The free body diagram with the nodes and members labelled is given below.



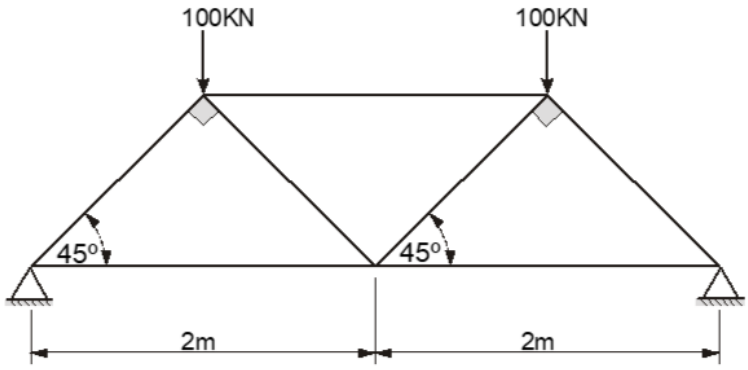
300kg x 9.8 = 2940N

i) In this type of problem where the structure is acting as a beam, find R1 and R_2 using moments. Take moments about R_1 -

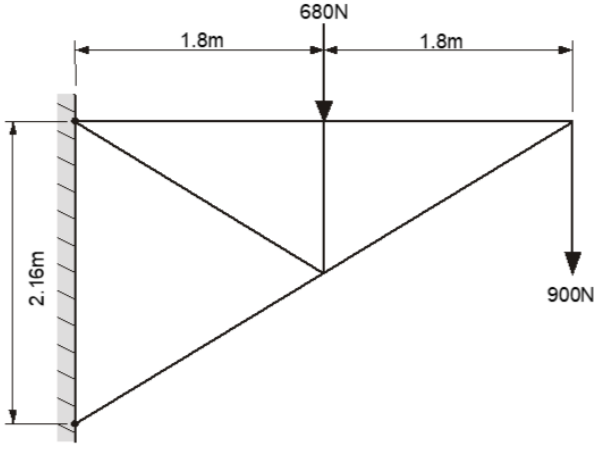
ii) Now complete this task and find R_1 and the forces in A, B and C.

14. Calculate the reactions, the magnitude and nature of the forces in the members of the frame structures shown below.

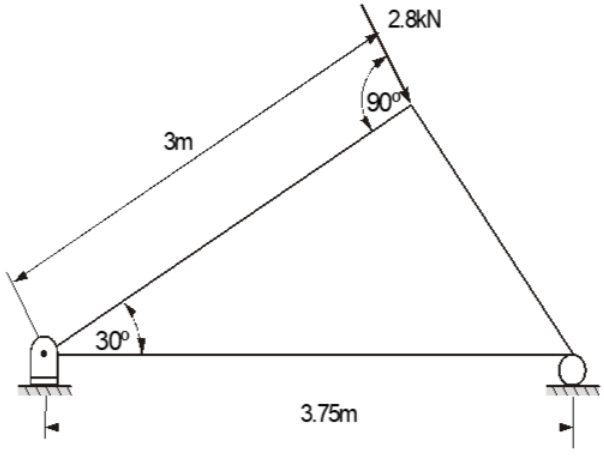
A)



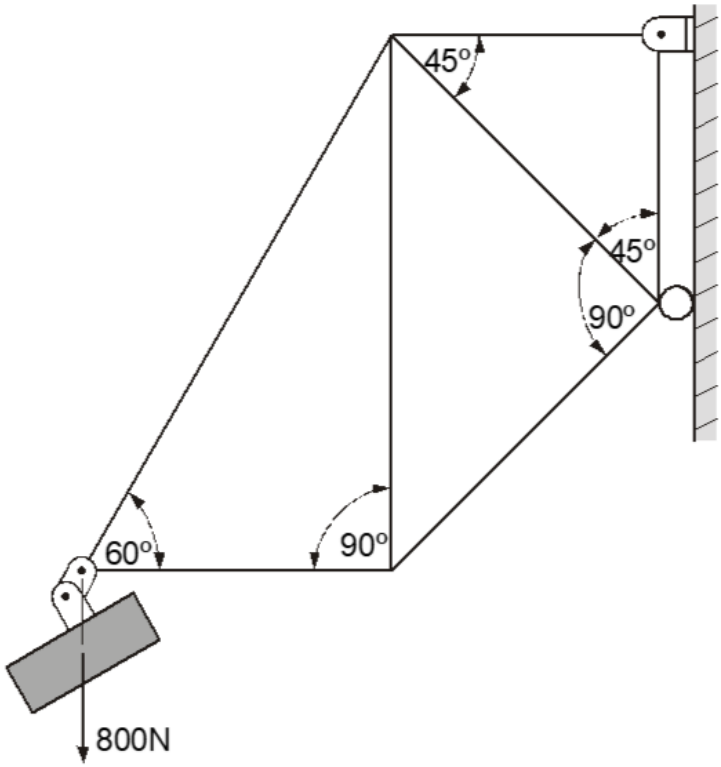
B)



C)



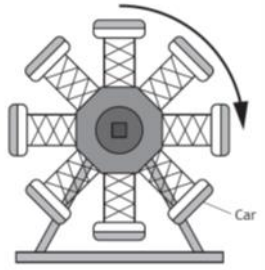
15. Calculate reactions and find the magnitude and nature of forces in the members in the lighting gantry structure shown below.



Calculating forces using simultaneous equations

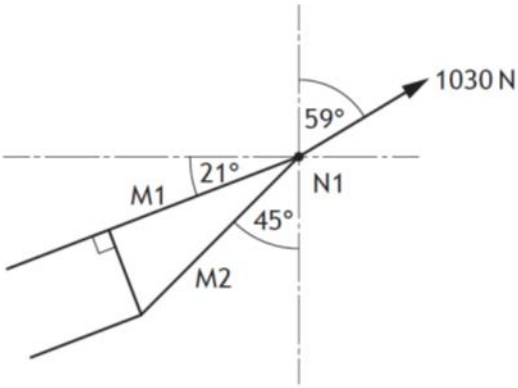
It is common to find nodes where there are two unknown quantities. In these situations, we need to use simultaneous equations.

Worked example.



A team of engineers are working on a new theme park ride.

As the ride spins a force of 1030N acts on a point of the structure as shown in the diagram below. Calculate, using simultaneous equations, the magnitude of forces in members M1 and M2.



Part 1: Sum of the Forces vertical and Horizontal.

Verticals. $\Sigma F_v = 0$

(F_v down = F_v up)

$M_1 \text{ Sin } 21^o + M_2 \text{ Sin} 45^o = 1030 \text{ Sin} 31^o$

$(M_1 \times 0.358) + (M_2 \times 0.707) = 530.49$ (Eqn 1)

Horizontals. $\Sigma F_H = 0$

$M_1 \text{ Cos} 21^o + M_2 \text{ Cos} 45^o = 1030 \text{ Cos} 31^o$

$(M_1 \times 0.934) + (M_2 \times 0.707) = 882.88$ (Eqn 2)

Place Eqn 1 into Eqn 2:

$M_1 \times 0.358 + M_2 \times 0.707 = 530.49$ (eqn 1)

$(M_1 \times 0.934) + (M_2 \times 0.707) = 882.88$ (Eqn 2)

Simplify Eqn 1 and Eqn 2 to get M_2 by itself:

$0.707M_2 = 530.49 - 0.358M_1$

$0.707M_2 = 882.88 - 0.934M_1$

The value of $0.707M_2$ cancels out and leaves:

$530.49 - 0.358M_1 = 882.88 - 0.934M_1$

Part 2: Calculation of M_1

$530.49 - 0.358M_1 = 882.88 - 0.934M_1$ (simplified equation)

$+0.934M_1 - 0.358M_1 = 882.88 - 530.49$ (rearrange equation)

$0.576M_1 = 352.39$

$M_1 = 352.39 \div 0.576$

$M_1 = 611.788 =$ 612N (now substitute into Eqn M_2)

Part 3: Calculation of M_2

$M_2 \times 0.707 = 530.49 - M_1 \times 0.358$

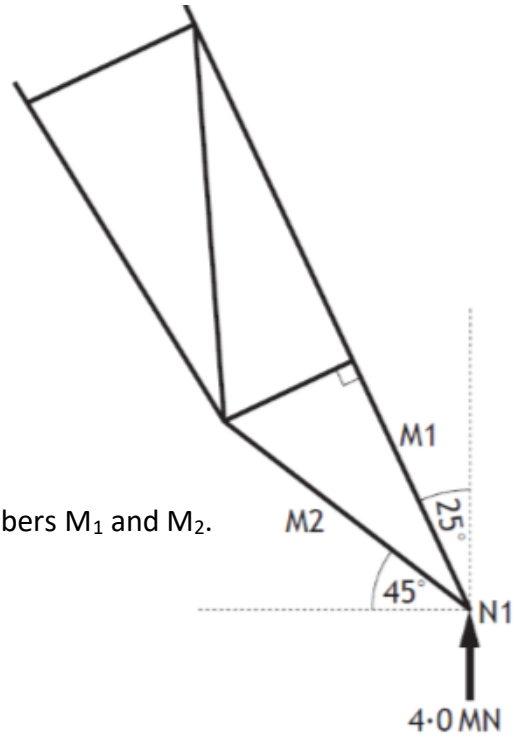
$M_2 = 530.49 - (612 \times 0.358) \div 0.707$

$M_2 =$ 440.0N

Nodal analysis with simultaneous equations

A new railway station is being built. A structural engineering company has been asked to produce detailed information about a possible station roof.

Detail from a proposed design for one of the roof's steel support legs is shown.



Node N1 is in static equilibrium. M_2 is a tie.

Calculate the magnitude of the forces in members M_1 and M_2.

$-M_2v + M_1v - 4.0 = 0$

$-M_2 \text{ Sin } (45^\circ) + M_1 \text{ Cos } (25^\circ) - 4.0 \times 106 = 0$

$-0{\cdot}707\ M_2 + 0{\cdot}906\ M_1 - 4.0 \times 106 = 0$ ***(eqn 1 now provided)***

$\Sigma Fh = 0$ (sum of the forces horizontal)

$M_2h - M_1h = 0$

$M_2 \text{ Cos } (45^\circ) - M_1 \text{ Sin } (25^\circ) = 0$

$0{\cdot}707\ M_2 - 0{\cdot}4225\ M_1 = 0$ (***eqn 2 now provided)***

(**Solve simultaneous equations by combining eqn 1 with eqn 2)**

$0{\cdot}906\ M_1 - 0{\cdot}4225\ M_1 - 4.0 \times 106 = 0$ (Notice 0.7070 M_2 cancels out from both equations)

$0{\cdot}484\ M_1 = 4.0 \times 106$

$M_1 = 8{\cdot}26 \times 106$ N = 8·3 MN (to 2 significant figures)

From Eqn 1, $M_2 = 0{\cdot}4225 \div 0{\cdot}707 \times 8{\cdot}26 = 4{\cdot}9$ MN (to 2 significant figures)

Additional worked examples

1)

The forces exerted on the windscreen-wiper arm by a retaining spring and by the windscreen wiper are shown in Figure Q3(*b*). P is the pivot-point.



Figure Q3(*b*) All dimensions in mm

Calculate, for a force, W, of 6·25 N:

(*a*) the force S exerted by the spring (take moments about P);

(*b*) the magnitude and direction of the reaction force at the pivot P.

a)

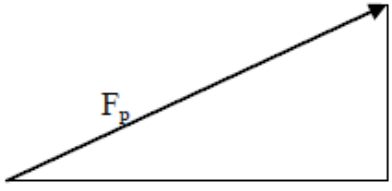
Taking moments about the pivot:

$\Sigma M_P = 0$ (c/w +ve)
$(S \cos 65° \times 120) - (6{\cdot}25 \sin 50° \times 500) = 0$ 2 for each term
$S = 47{\cdot}2$ N answer, including unit

b)

Taking horizontal components
$\Sigma F_H = 0$ ($\rightarrow$ +ve)
$F_{p(H)} + 6{\cdot}25 \cos 50° - 47.2 \sin 65° = 0$ 1 for each term
$F_{p(H)} = 38{\cdot}8$ N ($\rightarrow$) answer (unit not necessary)

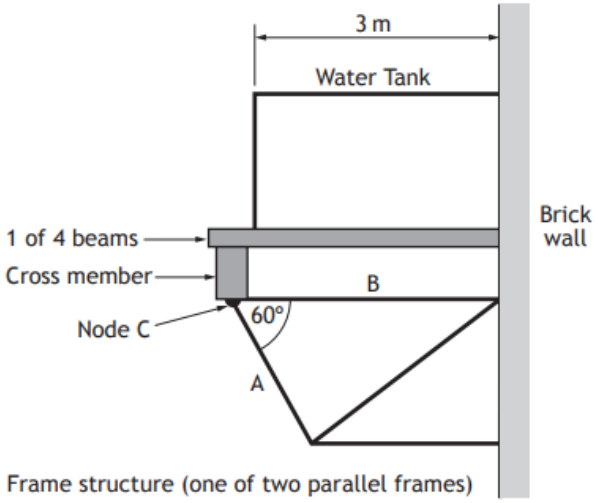
Taking vertical components:
$\Sigma F_V = 0$ ($\uparrow$+ve)
$F_{P(V)} + 6{\cdot}25 \sin 50° - 47{\cdot}2 \cos 65° = 0$ 1 for each term
$F_{P(V)} = 15{\cdot}2$ N ($\uparrow$) answer (unit not necessary)



$F_p^2 = 15{\cdot}2^2 + 38{\cdot}8^2$
$F_p = 41{\cdot}7$ kN (including unit)
$\text{Tan}^{-1}\ \theta = 15{\cdot}2/38{\cdot}8$
$\theta = 21{\cdot}4°$ (or 68.6°) answer 1
direction shown 1

2) The figure below shows one of the frame structures which support a tank. The force exerted by the water in the tank at node C is 58·8 KN.

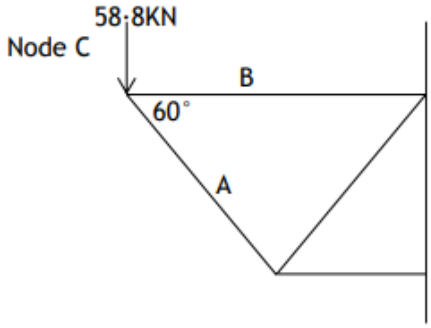
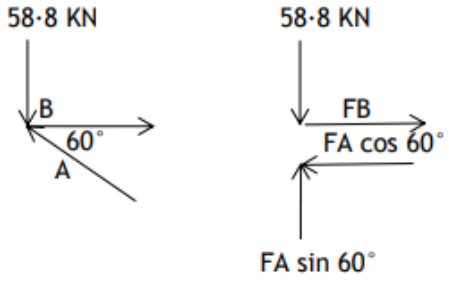
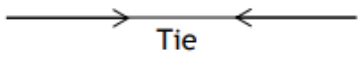
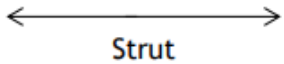
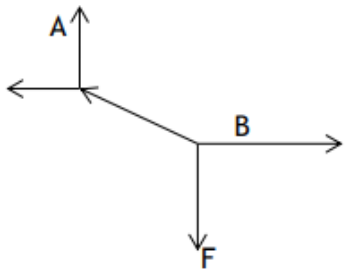
i) Calculate the magnitude and determine the nature of the forces in members labelled A and B.



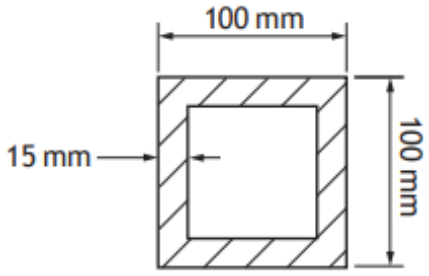
Frame structure (one of two parallel frames)

Side View

Answer

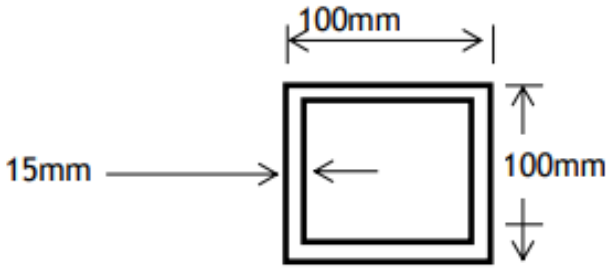
Expected Answer(s)	Max Mark	Additional Guidance
 Node C  ΣFv = 0 ΣFh = 0 58·8KN = FA sin 60° FB = FA cos 60° 58·8KN = FA × 0·866 FB = 67·9KN × 0·5 67·9 KN = FA (STRUT) FB= 33·95 KN(Tie) Answer and unit **(1)** Answer and unit**(1)** nature of BOTH members **(1)**	3	For nature tension/compression or force arrows on members or node acceptable   

Question 2 continued: Member A is a 100 mm × 100 mm square tubular section with a uniform wall thickness of 15 mm as shown below.



ii) Calculate the stress present in member A.

Answer

Expected Answer(s)	Max Mark	Additional Guidance
 δ=F/A Area 1 = 100 × 100 = 10000mm² Area 2 = 70 × 70 = 4900mm² **both values (1)** Total shaded area = A1 – A2 = 10000mm² – 4900mm² = 5100mm² **(1)** Force = 67900N (from b(i)) = 67900N ÷ 5100mm² = 13·3Nmm^{-2} answer and unit **(1)**	3	Units are not required for the mark for intermediate steps. Common error is using 85 × 85 as this internal size this gives an answer of 24·5 Nmm^{-2} for 2 marks

Exam style questions

1)

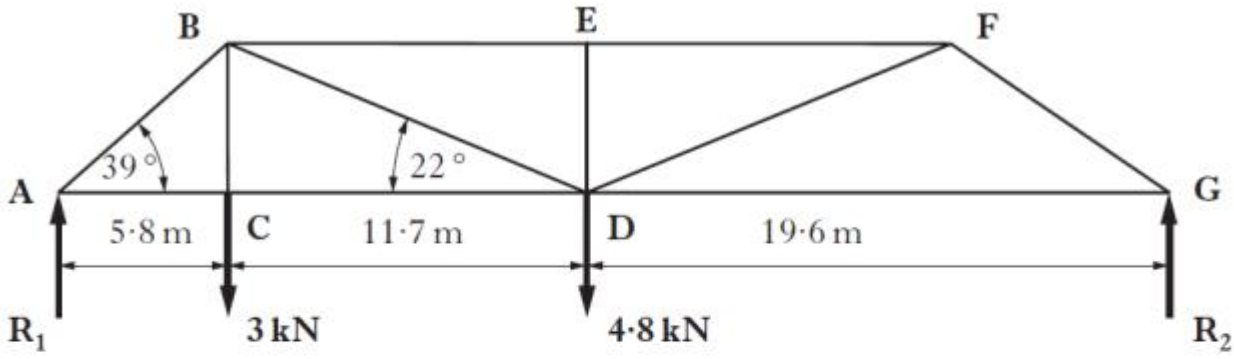


Figure Q9(*a*)

For the loading conditions shown in Figure Q9(*a*):

(*a*) calculate the reaction force R_1 acting at A;

(*b*) calculate the **magnitude** and **nature** of the forces in members AB, AC and BD.

2)

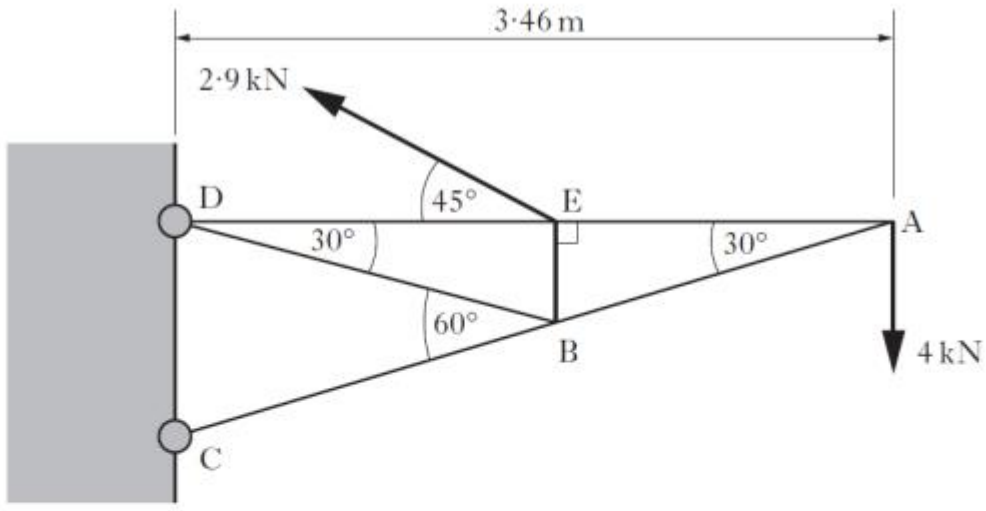


Figure Q9(*a*)

For a load of 4 kN acting at A:

(*a*) calculate the magnitude of the reaction force at C by taking moments about D;

(*b*) calculate the magnitude and nature of the forces in members AB, AE and BD.

3)

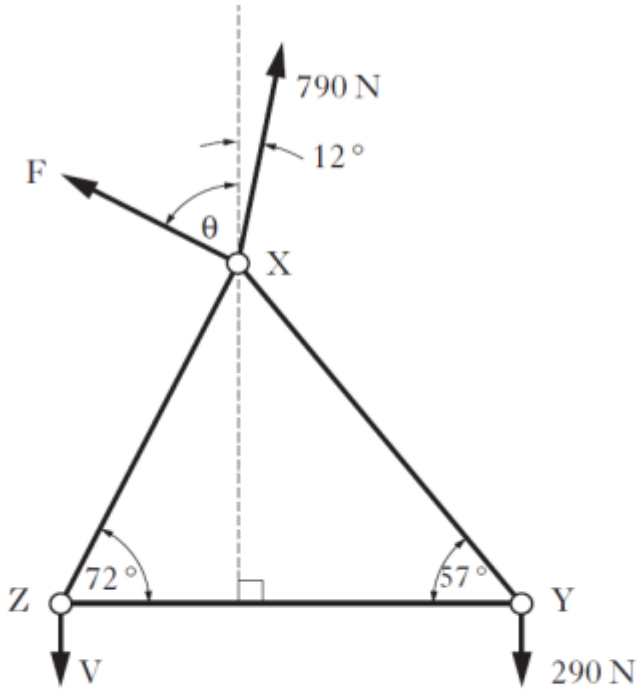


Figure Q7(*b*)

Calculate:

(i) the magnitude of the forces in members XY, YZ and XZ;

(ii) the magnitude of the forces in V and F;

(iii) the angle θ.